

Decomposing Modal Collapse: A Four-Valued Analysis of Gödel–Scott Positivity

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Abstract

I study the inferential architecture of the Gödel–Scott ontological theory under a bilateral four-valued semantics admitting both inconsistent and incomplete information. Scott’s theory is fixed as a classical control and reconstructed over a Belnap–Dunn/FDE kernel with bilateral Kripke modality, actualist individual quantification, and higher-order positivity. The original conjecture that the positive and negative modal-collapse channels separate fails for a structural reason: under involutive negation the two informative signed schemata are substitution instances of each other. The decomposition therefore occurs upstream. Scott’s A1 splits into two directional information flows with different jobs; two explicit S5 models satisfying the control stack, strong A1, and positivity rigidity refute positive T2 through separate glut and gap mechanisms; a targeted regularity package recovers it; and once T2 is available, symmetry alone suffices for positive T3.

Comparing the Scott, Anderson, and Fitting architectures then shows that these variants relocate the modal work rather than merely weakening the frame. The Fitting case is the sharpest. Identifying the extensional property domain with every bilateral extension makes the relevant consistency package incompatible with any positive Godlike witness, so I reconstruct the candidate over a selected FDE-negation-closed domain of admissible extensions without globally excluding gluts. On that domain the Godlikeness-as-essence route splits in two: a classification route driven by positivity completeness, and a quotient route that factors admissible extensions through positive-property profiles and leaves positivity gaps unresolved. The quotient is the fixed-point condition of a canonical closure operator, and a four-valued δ -filter on those fixed points preserves the gap exactly as long as ultrafilter maximality is withheld. Throughout, Lean 4.30.0 carries the general theorems and an executable finite-model oracle carries the countermodels and bounded searches, so unbounded claims and bounded evidence stay separate.

1 Introduction

Gödel’s ontological argument and Dana Scott’s well-known modification form an unusually compact laboratory for higher-order modal reasoning. Their interest for the present paper is structural rather than theological. The theory combines a primitive predicate of positive properties, modal rigidity, higher-order quantification, essence, and necessary existence, while its derivational behavior has been studied extensively with automated and interactive theorem provers [5, 4]. In Scott’s variant the main existence theorem is accompanied by the well-known phenomenon of *modal collapse*, schematically

$$\chi \rightarrow \Box\chi,$$

according to which every truth is necessary.

Recent machine-checked work sharpens the diagnosis of this phenomenon. In particular, the existence of an ultrafilter-like structure among positive properties does not by itself force collapse; rigidity of positivity is a principal modal ingredient [3]. This raises a natural question that classical two-valued semantics cannot express: which information flows inside positivity, essence, and necessary existence are actually responsible for reconstructing the collapse mechanism?

I study this question in a bilateral four-valued setting. Instead of assigning a single Boolean value to a formula, I track positive and negative support independently. A formula may therefore be supported only, opposed only, both supported and opposed, or neither. This gives the four values

$$\mathbf{T} = (1, 0), \quad \mathbf{F} = (0, 1), \quad \mathbf{B} = (1, 1), \quad \mathbf{N} = (0, 0),$$

following the Belnap–Dunn/FDE tradition [2, 6].

The central research question is:

Research Question 1.1 (Upstream structure of modal collapse). Under a four-valued bilateral modal semantics, which components of positivity–negation transfer, positivity rigidity, informational regularity, essence, and higher-order witness structure reconstruct the classical Gödel–Scott modal-collapse mechanism?

The informative signed collapse pair is

$$\text{MC}^+ : \quad +\chi \Rightarrow +\Box\chi, \quad \text{MC}^- : \quad -\chi \Rightarrow +\Box\neg\chi.$$

These are not independent in the fixed FDE setting. Since $\neg\chi$ is equivalent to $+\neg\chi$, MC^- is exactly MC^+ instantiated at $\neg\chi$, and conversely by involutive negation. The original collapse-separation conjecture is therefore falsified rather than preserved by an ad hoc semantic modification.

The decomposition instead occurs upstream. Scott’s A1 splits into two independent directional information flows,

$$A1_L : \quad -P(\varphi) \Rightarrow +P(\neg\varphi),$$

$$A1_R : \quad +P(\neg\varphi) \Rightarrow -P(\varphi).$$

The two directions acquire different jobs. $A1_R$, together with a semantic positive-support form of A2, yields possible exemplification for truth-only positive properties. $A1_L$ participates in the reflection and essence branch. A positivity glut blocks the unrestricted T1 analogue because the negative support produced by the classical reductio may already be present.

A natural bilateral reconstruction of Scott’s essence definition does not make Godlikeness an essence automatically. Two finite S5 models satisfy the earlier control axioms, strong A1, and positive rigidity while refuting positive T2: one fails through inconsistent exemplification and one through incomplete positivity information. A targeted regularity package recovers positive T2, after which the necessary-existence branch becomes comparatively robust. Positive support for A5 suffices for a positive T3 in S5, even if necessary existence itself is glutty.

Once positive T2 and T3 are available, modal collapse follows directly from essence and constant-property embedding. The central non-classical difficulty is therefore no longer modal collapse itself, nor even T3. It is the weakest principled recovery of Godlikeness-as-essence from the lower-level four-valued Scott machinery.

Gate 7 machine-checks this dependency picture. Lean 4.30.0 verifies the collapse-channel equivalence, the truth-only T1 theorem, the sufficient T2 recovery theorem, conditional positive T3, witness supply, the essence-compressed positive-collapse theorem, and bilateral classical-recovery interfaces. An executable finite-model layer retains the glut/gap countermodels and establishes bounded individual indispensability of the four current T2 recovery assumptions in a complete generated two-world, one-entity family. The remaining T2-minimality question is therefore explicitly global rather than a claim inferred from bounded search.

The contribution sought is not a “stronger proof of God”. The intended contribution is an explanatory dependency analysis showing where four-valued inconsistency and incompleteness alter the inferential architecture of the Scott theory, including negative results and countermodels that locate hidden classical regularity assumptions.

Methodological discipline. The development proceeds conservatively. I freeze a classical Scott baseline, fix the propositional and modal semantics, and then lift individual Gödel–Scott principles only when their semantic prerequisites are explicit. Failed conjectures are retained as results rather than repaired by changing the semantics after the fact. Implication-sensitive

clauses are represented by signed semantic entailment relations rather than by prematurely choosing a four-valued object-language conditional.

Status of this manuscript. Version 0.19 reports Gates 0–12. Beyond the semantic reconstruction through T3 and modal collapse, it records the Lean 4.30.0 development, the executable countermodels and bounded searches, the Anderson and Fitting comparisons, the positive-profile quotient with its closure operator and explicit quotient type, a first four-valued filter layer on the resulting fixed points, and the paired-neighborhood representation of the relational modal control. Global T2 minimality, theorem-specific transport to non-principal neighborhood frames, and the line-by-line correspondence audit against the published HOL developments remain open.

2 Classical Scott Baseline

STATUS: ESTABLISHED AT THE SEMANTIC LEVEL.

I use Scott’s variant as the classical control theory, following the modern formal presentation of Benzmüller and Scott [5]. The reference environment uses intensional properties over possible worlds, actualist quantification over individuals, possibilist quantification over properties, and an S5 reference frame. Later experiments may weaken the modal frame, but such changes are not part of the baseline.

Let $P(\varphi)$ mean that property φ is positive. The relevant axioms and definitions are:

$$\text{A1} \quad \neg P(\varphi) \leftrightarrow P(\neg\varphi), \quad (1)$$

$$\text{A2} \quad P(\varphi) \wedge \Box \forall^E y (\varphi(y) \rightarrow \psi(y)) \rightarrow P(\psi), \quad (2)$$

$$\text{D1} \quad G(x) \equiv \forall \varphi (P(\varphi) \rightarrow \varphi(x)), \quad (3)$$

$$\text{A3} \quad P(G), \quad (4)$$

$$\text{A4} \quad P(\varphi) \rightarrow \Box P(\varphi), \quad (5)$$

$$\text{D2} \quad \varphi \text{ Ess } x \equiv \varphi(x) \wedge \forall \psi (\psi(x) \rightarrow \Box \forall^E y (\varphi(y) \rightarrow \psi(y))), \quad (6)$$

$$\text{D3} \quad NE(x) \equiv \forall \varphi (\varphi \text{ Ess } x \rightarrow \Box \exists^E y \varphi(y)), \quad (7)$$

$$\text{A5} \quad P(NE). \quad (8)$$

The standard theorem chain includes

$$\text{T1} \quad P(\varphi) \rightarrow \Diamond \exists^E x \varphi(x), \quad (9)$$

$$\Diamond \exists^E x G(x), \quad (10)$$

$$\text{T2} \quad G(x) \rightarrow G \text{ Ess } x, \quad (11)$$

$$\text{T3} \quad \Box \exists^E x G(x), \quad (12)$$

$$\text{MC} \quad \chi \rightarrow \Box \chi. \quad (13)$$

For the present project, T3 is a control theorem while MC is the main structural target. A proposed four-valued theory must state precisely in what sense this classical system is recovered when all relevant valuations are restricted to the consistent and complete values **T** and **F**.

The project-level baseline, including quantifier conventions and dependency notes, is maintained in `docs/SCOTT_BASELINE.md`.

3 Four-Valued Propositional Kernel

STATUS: ESTABLISHED AT THE SEMANTIC LEVEL.

The propositional base follows the information structure of Belnap–Dunn First-Degree Entailment [2, 6]. Every formula φ receives two independent bits

$$v(\varphi) = (t(\varphi), f(\varphi)) \in \{0, 1\}^2,$$

where $t(\varphi) = 1$ represents positive support and $f(\varphi) = 1$ negative support. I write $+\varphi$ and $-\varphi$ for these two satisfaction conditions.

Value	Pair	Reading
T	(1, 0)	positive support only
F	(0, 1)	negative support only
B	(1, 1)	both positive and negative support
N	(0, 0)	neither positive nor negative support

Atoms receive positive and negative support independently. The connectives are fixed by the bilateral clauses

$$\begin{array}{ll}
+\neg\varphi \iff -\varphi, & -\neg\varphi \iff +\varphi, \\
+(\varphi \wedge \psi) \iff +\varphi \wedge +\psi, & -(\varphi \wedge \psi) \iff -\varphi \vee -\psi, \\
+(\varphi \vee \psi) \iff +\varphi \vee +\psi, & -(\varphi \vee \psi) \iff -\varphi \wedge -\psi.
\end{array}$$

Positive semantic consequence preserves positive satisfaction:

$$\Gamma \models_{\text{FDE}} \varphi$$

iff every valuation positively satisfying every member of Γ also positively satisfies φ . Equivalently, the positively designated values are **{T, B}**. Negative satisfaction remains separately observable and will be used for bilateral meta-conditions in later sections.

Crucially, no object-language implication is fixed at this stage. Although $\neg\varphi \vee \psi$ is definable, treating it as *the* conditional would prematurely determine the lifting of A2, D1, D2, and D3.

Proposition 3.1 (Paraconsistency). *The kernel is non-explosive: $p, \neg p \not\models_{\text{FDE}} q$.*

Proof. Let $v(p) = \mathbf{B}$ and $v(q) = \mathbf{N}$. Then p and $\neg p$ are both positively satisfied, while q is not. $\square$

Proposition 3.2 (Paracompleteness). *Excluded middle is not valid: $\not\models_{\text{FDE}} p \vee \neg p$.*

Proof. Let $v(p) = \mathbf{N}$. Negation preserves $\mathbf{N}$, so neither disjunct is positively satisfied and hence neither is their disjunction. $\square$

Proposition 3.3 (Classical recovery of the propositional fragment). *If atomic valuations are restricted to $\{\mathbf{T}, \mathbf{F}\}$, then the clauses for $\neg$, $\wedge$, and $\vee$ remain inside $\{\mathbf{T}, \mathbf{F}\}$ and coincide with the classical truth tables.*

Proof. Immediate by structural induction on formulas using the bilateral clauses above. $\square$

The full gate specification is maintained in `docs/FOUR_VALUED_KERNEL.md`.

4 Four-Valued Modal Semantics

STATUS: ESTABLISHED AT THE SEMANTIC LEVEL.

The modal layer is fixed in two stages. For the first Gödel–Scott decomposition experiments I use a conservative bilateral Kripke semantics as a control. I then represent this semantics inside a paired-neighborhood framework that can later be generalized without changing the four-valued propositional kernel. This separates two possible sources of non-classical behavior: four-valued information and non-Kripke modal structure.

4.1 Relational control semantics

Let $\mathcal{F} = (W, R)$ be a Kripke frame. Positive and negative satisfaction for necessity are defined by

$$w \models^+ \Box \varphi \iff \forall v (wRv \Rightarrow v \models^+ \varphi),$$

$$w \models^- \Box \varphi \iff \exists v (wRv \wedge v \models^- \varphi).$$

Possibility uses the converse quantifier pattern:

$$w \models^+ \Diamond \varphi \iff \exists v (wRv \wedge v \models^+ \varphi),$$

$$w \models^- \Diamond \varphi \iff \forall v (wRv \Rightarrow v \models^- \varphi).$$

These clauses belong to the established family of Belnap–Dunn modal semantics [11, 14]; recent work also develops an S5-based four-valued setting for essence and accident modalities [12].

Proposition 4.1 (Bilateral modal duality). *Under the relational clauses,*

$$\Diamond\varphi \equiv \neg\Box\neg\varphi \quad \text{and} \quad \Box\varphi \equiv \neg\Diamond\neg\varphi$$

hold at both positive and negative satisfaction levels.

Proof. For example,

$$\begin{aligned} w \models^+ \neg\Box\neg\varphi &\iff w \models^- \Box\neg\varphi \\ &\iff \exists v (wRv \wedge v \models^- \neg\varphi) \\ &\iff \exists v (wRv \wedge v \models^+ \varphi) \\ &\iff w \models^+ \Diamond\varphi. \end{aligned}$$

The remaining signed equivalences are analogous. $\square$

Proposition 4.2 (Classical recovery of the relational modal fragment). *If atomic valuations are restricted to $\{\mathbf{T}, \mathbf{F}\}$, then formulas generated with $\neg, \wedge, \vee, \Box, \Diamond$ remain in $\{\mathbf{T}, \mathbf{F}\}$, and the signed modal clauses coincide with ordinary Kripke semantics.*

Proof. The propositional step is established in Section 3. For a classically valued formula φ , positive and negative satisfaction are complementary at every world. Hence $\Box\varphi$ is positively satisfied exactly when every accessible world positively satisfies φ , and negatively satisfied exactly when some accessible world negatively satisfies it. Exactly one of these conditions holds. The diamond case is dual. $\square$

If R is an equivalence relation, the classical fragment therefore recovers the S5 environment used as the Scott control setting.

4.2 Modal aggregation of gluts and gaps

The signed quantifier pattern retains the full four-valued range. For $\Box\varphi$, value $\mathbf{B}$ occurs when every accessible world positively supports φ while at least one also negatively supports it; value $\mathbf{N}$ occurs when positive support fails at some accessible world while negative support occurs nowhere. Dually, $\Diamond\varphi = \mathbf{B}$ when positive support occurs somewhere and negative support occurs everywhere, while $\Diamond\varphi = \mathbf{N}$ when positive support occurs nowhere and negative support fails somewhere.

Thus modalization does not erase the distinction between inconsistency and incompleteness.

4.3 Paired-neighborhood representation

For a formula φ , write

$$\llbracket\varphi\rrbracket^+ = \{w \in W \mid w \models^+ \varphi\}, \quad \llbracket\varphi\rrbracket^- = \{w \in W \mid w \models^- \varphi\}.$$

At each world introduce two families $\mathcal{U}_w, \mathcal{H}_w \subseteq \mathcal{P}(W)$, interpreted provisionally by their modal roles: $\mathcal{U}$ is a universal/core family and $\mathcal{H}$ a hit/witness family. Define

$$\begin{aligned} w \models^+ \Box \varphi &\iff \llbracket \varphi \rrbracket^+ \in \mathcal{U}_w, & w \models^- \Box \varphi &\iff \llbracket \varphi \rrbracket^- \in \mathcal{H}_w, \\ w \models^+ \Diamond \varphi &\iff \llbracket \varphi \rrbracket^+ \in \mathcal{H}_w, & w \models^- \Diamond \varphi &\iff \llbracket \varphi \rrbracket^- \in \mathcal{U}_w. \end{aligned}$$

The crosswise use of the two families preserves the same bilateral box/diamond duality for arbitrary $\mathcal{U}$ and $\mathcal{H}$.

Every relational frame canonically induces such a pair by

$$\begin{aligned} \mathcal{U}_w^R &:= \{X \subseteq W \mid R[w] \subseteq X\}, \\ \mathcal{H}_w^R &:= \{X \subseteq W \mid R[w] \cap X \neq \emptyset\}. \end{aligned}$$

Proposition 4.3 (Relational representation). *The paired-neighborhood semantics induced by $\mathcal{U}^R, \mathcal{H}^R$ agrees exactly with the relational bilateral semantics at both signed satisfaction levels.*

Proof. Immediate from the definitions: membership in $\mathcal{U}_w^R$ is exactly universal coverage of $R[w]$, while membership in $\mathcal{H}_w^R$ is exactly nonempty intersection with $R[w]$. $\square$

This establishes a hierarchy rather than a competition between Kripke and neighborhood semantics. The relational control model is a principal paired-neighborhood model.

Gate 11 machine-checks this representation in Lean without a finiteness or frame-condition premise. The formal module defines the four paired clauses, proves both signed box/diamond dualities under FDE negation, and shows that the canonical construction from R is definitionally identical to the existing relational clauses. An exhaustive two-world regression independently compares box and diamond for all sixteen relations, both source worlds, and all sixteen bilateral valuations: all 1024 signed-value operator comparisons agree.

4.4 Classical recovery beyond relational frames

For arbitrary paired neighborhoods, modalization may itself turn classically valued input into **B** or **N**. To preserve classical bivalence impose the complement-duality condition

$$X \in \mathcal{U}_w \iff W \setminus X \notin \mathcal{H}_w.$$

If φ is classical-valued, then

$$\llbracket \varphi \rrbracket^- = W \setminus \llbracket \varphi \rrbracket^+,$$

so the condition guarantees that exactly one signed clause for $\Box\varphi$ holds, and likewise for $\Diamond\varphi$. Every relationally induced pair satisfies this condition.

The two-world local neighborhood space makes the inclusions strict. Among all 256 pairs $(\mathcal{U}_w, \mathcal{H}_w)$, exactly 16 are complement-dual, only 4 are principal relational, and hence 12 are non-principal while preserving classical modal bivalence. Across arbitrary non-dual pairs, classically valued inputs realize all four outputs **T**, **F**, **B**, **N**. Thus non-principality and failure of complement duality are independent experimental changes: the former can preserve bivalence, whereas the latter can make modalization itself introduce a glut or a gap.

Experimental decision. For Gates 3–8 I keep the S5 relational control semantics fixed. This makes the separation analysis attributable to the bilateral lifting of Gödel–Scott positivity rather than to a simultaneous weakening of modal structure. Gate 11 now opens the second experimental axis only after that dependency map is established. The representation and strict bounded hierarchy are complete; transporting selected higher-order Scott, Anderson, and Fitting theorems to non-principal frames remains separate work.

The full Gate-2 specification is maintained in `docs/MODAL_LIFT.md`.

5 Bilateral Positivity

STATUS: ESTABLISHED AT THE SEMANTIC LEVEL.

The positivity predicate P is the central non-logical primitive of the Scott theory. In the four-valued reconstruction, statements of the form $P(\varphi)$ carry positive and negative information independently.

5.1 Directional decomposition of A1

Scott’s A1 is

$$\neg P(\varphi) \leftrightarrow P(\neg\varphi).$$

A split into two biconditional channel schemas would be redundant: because property negation is involutive and A1 is a schema over all properties, substituting $\neg\varphi$ for φ converts either biconditional into the other. I therefore decompose A1 by direction:

$$A1_L : \quad \neg P(\varphi) \Rightarrow +P(\neg\varphi),$$

$$A1_R : \quad +P(\neg\varphi) \Rightarrow \neg P(\varphi).$$

Here and below $\Rightarrow$ remains a metalanguage implication.

Proposition 5.1 (Directional independence of A1). *Neither $A1_L$ nor $A1_R$ semantically entails the other in the four-valued positivity semantics.*

Proof. For $A1_L$ without $A1_R$, let $v(P(\varphi)) = \mathbf{T}$ and $v(P(\neg\varphi)) = \mathbf{N}$. The antecedents of all $A1_L$ instances on this property pair lack negative support, while the mirrored $A1_R$ instance from $+P(\varphi)$ to $-P(\neg\varphi)$ fails. Conversely, let $v(P(\varphi)) = \mathbf{F}$ and $v(P(\neg\varphi)) = \mathbf{N}$. Then the $A1_R$ antecedents lack positive support, whereas $-P(\varphi)$ holds without $+P(\neg\varphi)$, so $A1_L$ fails. $\square$

When both directions hold, strong bilateral A1 yields

$$+P(\neg\varphi) \Leftrightarrow -P(\varphi)$$

and, by substitution,

$$-P(\neg\varphi) \Leftrightarrow +P(\varphi).$$

Thus

$$v(P(\neg\varphi)) = \text{swap}(v(P(\varphi))),$$

so property negation exchanges $\mathbf{T}$ and $\mathbf{F}$ while preserving $\mathbf{B}$ and $\mathbf{N}$.

5.2 A4 and informative negative rigidity

Scott's A4,

$$P(\varphi) \rightarrow \Box P(\varphi),$$

has the direct positive-support lifting

$$R^+ : \quad +P(\varphi) \Rightarrow +\Box P(\varphi).$$

A previously considered negative clause,

$$-P(\varphi) \Rightarrow -\Box P(\varphi),$$

is not informative on the fixed S5 control frames. By the bilateral modal clause, $-\Box\psi$ merely requires some accessible world to negatively support ψ ; reflexivity makes the current world such a witness whenever $-\psi$ already holds.

The informative negative-persistence condition is instead

$$R_{\text{nec}}^- : \quad -P(\varphi) \Rightarrow +\Box\neg P(\varphi),$$

equivalently

$$-P(\varphi) \Rightarrow -\Diamond P(\varphi).$$

Proposition 5.2 (A1 couples the rigidity channels). *Globally assuming $A1_L$, $A1_R$, and R^+ entails R_{nec}^- .*

Proof. Suppose $-P(\varphi)$ at a world w . By $A1_L$, $+P(\neg\varphi)$. Applying R^+ to $\neg\varphi$ gives $+\Box P(\neg\varphi)$, so every accessible world positively supports $P(\neg\varphi)$. Global $A1_R$ therefore makes every accessible world negatively support $P(\varphi)$. By the bilateral box clause this is exactly $+\Box\neg P(\varphi)$. $\square$

This proposition is the first direct evidence that A1 is not merely a static complement axiom. In the presence of A4 it transfers modal persistence from the positive information channel into the negative one.

5.3 A local obstruction to classical positivity reflection

For the classical collapse spine it is useful to isolate the minimal support interface

$$D1^+ : \quad +G(x) \wedge +P(\varphi) \Rightarrow +\varphi(x).$$

This is not yet the final four-valued definition of Godlikeness.

The classical proof uses a local reflection move

$$G(x), Z(x) \Rightarrow P(Z).$$

Its signed analogue would be

$$REF^+ : \quad +G(x) \wedge +Z(x) \Rightarrow +P(Z).$$

Even strong A1 together with $D1^+$ does not entail REF^+ .

A glut counterassignment takes

$$v(Z(x)) = \mathbf{B}, \quad v(P(Z)) = \mathbf{F}, \quad v(P(\neg Z)) = \mathbf{T}.$$

Strong A1 is respected, and $D1^+$ requires positive support for $\neg Z(x)$, equivalently negative support for $Z(x)$, which the glut already supplies. Yet $+P(Z)$ fails.

A distinct gap counterassignment takes

$$v(Z(x)) = \mathbf{T}, \quad v(P(Z)) = v(P(\neg Z)) = \mathbf{N}.$$

Strong A1 preserves the positivity gap, $D1^+$ is silent, and again $+P(Z)$ fails.

These are local signed counterassignments to the proof interface, not yet full higher-order countermodels. They expose two different classical assumptions hidden in the ordinary reductio: consistency of the relevant exemplification and completeness of the relevant positivity information. Both dimensions will therefore be tracked explicitly in the collapse analysis.

6 From Positivity Rigidity to Modal Collapse

The classical proof structure identifies a narrow pressure point for the four-valued analysis. Fix a world w , an individual x , and a property Z . In the Scott theory,

$$G(x)@w, Z(x)@w$$

forces $P(Z)@w$: if Z were non-positive, A1 would make $\neg Z$ positive, while D1 would then force the God-like x to exemplify $\neg Z$, contradicting $Z(x)$. A4 then yields $\Box P(Z)$, and D1 propagates Z to God-like entities at accessible worlds. Thus the local classical spine is

$$G(x), Z(x) \xrightarrow{A1+D1} P(Z) \xrightarrow{A4} \Box P(Z) \xrightarrow{D1} \Box \forall z (G(z) \rightarrow Z(z)).$$

Substituting the constant-in-individuals property $Z := \lambda z.\chi$ embeds an arbitrary proposition χ into this mechanism. Together with T3, which supplies necessarily existing God-like witnesses, and suitable frame conditions, the standard Scott setting yields

$$\chi \rightarrow \Box\chi.$$

The detailed classical dependency analysis is maintained in `docs/MODAL_COLLAPSE_SPINE.md`. This local structure is consonant with the recent machine-checked diagnosis that positivity rigidity, rather than ultrafilter structure alone, is a principal driver of collapse [3].

6.1 Correct bilateral collapse targets

The informative signed persistence principles are

$$\text{MC}^+ : \quad +\chi \Rightarrow +\Box\chi$$

and

$$\text{MC}^- : \quad -\chi \Rightarrow +\Box\neg\chi.$$

The earlier candidate $-\chi \Rightarrow -\Box\chi$ is discarded because it follows automatically on reflexive frames: the current world itself witnesses negative satisfaction of $\Box\chi$.

Theorem 6.1 (Collapse-channel equivalence). *Under the fixed FDE negation, with both schemata ranging over all formulas,*

$$\text{MC}^+ \iff \text{MC}^-.$$

Proof. Assume MC^+ and $-\chi$. By FDE negation, $+\neg\chi$. Instantiating MC^+ with $\neg\chi$ gives $+\Box\neg\chi$, hence MC^- .

Conversely, assume MC^- and $+\chi$. Then $-\neg\chi$. Instantiating MC^- with $\neg\chi$ yields $+\Box\neg\neg\chi$, which is equivalent to $+\Box\chi$ because FDE negation is involutive. $\square$

Thus the original conjecture that the informative positive and negative collapse channels might separate is falsified for the present semantics. The decomposition must be sought upstream rather than forced at the collapse schema itself.

6.2 Rigidity separates before A1 couples it

Write

$$R^+ : \quad +P(\varphi) \Rightarrow +\Box P(\varphi)$$

and

$$R_{\text{nec}}^- : \quad -P(\varphi) \Rightarrow +\Box\neg P(\varphi).$$

Without A1 these principles are independent. On the two-world S5 frame with universal accessibility, assign a property ψ the positivity values

$$v_{w_0}(P(\psi)) = \mathbf{N}, \quad v_{w_1}(P(\psi)) = \mathbf{F}.$$

If all other relevant positivity statements receive $\mathbf{N}$, then R^+ holds vacuously while R_{nec}^- fails at w_1 . Replacing $\mathbf{F}$ by $\mathbf{T}$ yields the converse separation: R_{nec}^- holds vacuously while R^+ fails.

Strong A1 destroys this independence.

Proposition 6.2 (Rigidity coupling under strong A1). *Under $A1_L + A1_R$,*

$$R^+ \iff R_{\text{nec}}^-.$$

Proof. The direction $R^+ \Rightarrow R_{\text{nec}}^-$ was established in the positivity analysis. For the converse, suppose $+P(\varphi)$. By $A1_R$, instantiated at $\neg\varphi$, one obtains $-P(\neg\varphi)$. Applying R_{nec}^- to $\neg\varphi$ yields $+\Box\neg P(\neg\varphi)$. At each accessible world, $A1_L$ converts $-P(\neg\varphi)$ back into $+P(\varphi)$. Hence $+\Box P(\varphi)$. $\square$

The genuine bilateral separation therefore occurs at the level of positivity rigidity only in the absence of the full A1 coupling.

6.3 Minimal recovery of the local reflection bridge

The signed reflection target is

$$REF^+ : \quad +G(x) \wedge +Z(x) \Rightarrow +P(Z).$$

Strong A1 plus the minimal support interface

$$D1^+ : \quad +G(x) \wedge +P(\varphi) \Rightarrow +\varphi(x)$$

does not suffice: the glut and gap assignments analyzed in Section 5 block the classical reductio in different ways.

Introduce two explicit metalanguage regularity conditions for the relevant witness and property:

$$CONS_G : \quad +G(x) \wedge +Z(x) \Rightarrow \text{not } -Z(x),$$

and

$$COMP_P : \quad +P(Z) \text{ or } -P(Z).$$

Proposition 6.3 (Local reflection recovery). *At the abstract Gate-4 interface,*

$$A1_L + D1^+ + CONS_G + COMP_P \models REF^+.$$

Proof. Assume $+G(x)$ and $+Z(x)$. By $COMP_P$, either $+P(Z)$ or $-P(Z)$. In the latter case, $A1_L$ yields $+P(\neg Z)$, and $D1^+$ then gives $+\neg Z(x)$, equivalently $-Z(x)$. This contradicts $CONS_G$. Hence the negative-positivity alternative is excluded and $+P(Z)$ remains. $\square$

The package is locally minimal relative to this interface vocabulary. Removing $A1_L$, $D1^+$, $CONS_G$, or $COMP_P$ respectively admits a counterassignment while the other three remain satisfiable. The glut counterassignment witnesses the need for $CONS_G$; the gap counterassignment witnesses the need for $COMP_P$.

6.4 Conditional collapse kernel

Two higher-order interfaces are kept explicit because the final quantifier and Godlikeness semantics have not yet been frozen:

- GW : every world has some positively supported God-like witness;
- $CONST$: every modal formula χ is representable by the constant intensional property $Z_\chi := \lambda x.\chi$, preserving both signed satisfaction conditions.

Theorem 6.4 (Interface-level collapse package). *Assume globally*

$$A1_L + D1^+ + CONS_G + COMP_P + R^+ + GW + CONST.$$

Then MC^+ , and therefore also MC^- .

Proof. Suppose $+\chi$ at w . By GW , choose x with $+G(x)$ at w . By $CONST$, $+Z_\chi(x)$. The local reflection proposition yields $+P(Z_\chi)$. By R^+ , $+\Box P(Z_\chi)$.

Let v be accessible from w . Then $+P(Z_\chi)$ at v . By GW , choose y with $+G(y)$ at v . By $D1^+$, $+Z_\chi(y)$, and by $CONST$, $+\chi$ at v . Since v was arbitrary, $+\Box\chi$ at w . $\square$

This is deliberately an interface-level theorem rather than a theorem of the complete four-valued Gödel–Scott theory. Gate 5 must determine whether the final treatment of A2, A3, D1, and higher-order quantification justifies GW and $CONST$, and whether the regularity assumptions can be weakened or derived rather than postulated.

Gate-4 conclusion. The strongest current structural claim is therefore not that modal collapse itself splits. Under symmetric FDE negation it does not. The non-classical decomposition occurs upstream: positivity rigidity separates before strong A1 recouples it, while the classical reflection bridge additionally relies on consistency and completeness conditions that gluts and gaps defeat independently.

7 Reconstructing the Gödel–Scott Chain

Gates 5 and 6 reconstruct the higher-order existence branch through A2, A3, Godlikeness, essence, necessary existence, A5, and a positive analogue of Scott’s T3. The reconstruction keeps the object-language four-valued conditional unfixed; implication-sensitive clauses are represented semantically.

7.1 Actualist quantification and necessary positive entailment

I retain Scott's actualist quantification over individuals using a classical metalanguage existence relation $E(w, x)$. Positive and negative existential satisfaction are

$$\begin{aligned} w \models^+ \exists^E x \varphi(x) &\iff \exists x (E(w, x) \wedge w \models^+ \varphi(x)), \\ w \models^- \exists^E x \varphi(x) &\iff \forall x (E(w, x) \Rightarrow w \models^- \varphi(x)), \end{aligned}$$

with the dual universal clauses.

Necessary positive entailment is the semantic relation

$$\text{NEnt}_E^+(\varphi, \psi, w),$$

which holds iff at every world accessible from w , every actually existing positive instance of φ is a positive instance of ψ . The control lifting of A2 is

$$A2^+ : \quad +P(\varphi) \wedge \text{NEnt}_E^+(\varphi, \psi) \Rightarrow +P(\psi).$$

For Gate 6 I also use negative evidence against necessary entailment:

$$\text{NEnt}_E^-(\varphi, \psi, w)$$

iff some accessible actually existing object positively exemplifies φ and negatively exemplifies ψ . On the classical fragment these two relations reproduce truth and falsity of the necessary-entailment condition.

7.2 Truth-only positivity and possible exemplification

Define

$$\text{Pos}_T(\varphi) \iff +P(\varphi) \text{ and not } -P(\varphi),$$

where “not” is metalanguage denial.

Theorem 7.1 (Truth-only positive properties are possibly exemplified). *Assume globally $A1_R$ and $A2^+$. Then*

$$\text{Pos}_T(\varphi) \Rightarrow +\Diamond \exists^E x \varphi(x).$$

The proof is the Gate-5 reductio: failure of positive possible exemplification makes $\text{NEnt}_E^+(\varphi, \neg\varphi)$ vacuously true; A2 then gives $+P(\neg\varphi)$, and $A1_R$ yields $-P(\varphi)$, contradicting truth-only positivity. A one-world model with an empty actual domain and gluttony positivity shows that the stronger antecedent $+P(\varphi)$ is insufficient.

Scott A3 is therefore split into

$$A3^+ : \quad +P(G), \quad A3_T : \quad \text{Pos}_T(G).$$

The control branch uses $A3_T$, obtaining

$$\boxed{+\Diamond \exists^E x G(x)}.$$

7.3 Support-based Godlikeness

The control definition $G\text{-sup-v0.1}$ is

$$\begin{aligned} w \models^+ G(x) &\iff \forall \varphi (w \models^+ P(\varphi) \Rightarrow w \models^+ \varphi(x)), \\ w \models^- G(x) &\iff \exists \varphi (w \models^+ P(\varphi) \wedge w \models^- \varphi(x)). \end{aligned}$$

This recovers Scott D1 on the classical fragment, discharges the former $D1^+$ interface, and permits all four information states for Godlikeness. Full higher-order lambda abstraction supplies the constant-property interface $CONST$.

7.4 Bilateral essence

The Gate-6 control definition $Ess\text{-sup-v0.1}$ is

$$\begin{aligned} w \models^+ (\varphi \text{Ess } x) &\iff w \models^+ \varphi(x) \\ &\quad \wedge \forall \psi (w \models^+ \psi(x) \Rightarrow \text{NEnt}_E^+(\varphi, \psi, w)), \end{aligned}$$

while

$$\begin{aligned} w \models^- (\varphi \text{Ess } x) &\iff w \models^- \varphi(x) \\ &\quad \vee \exists \psi (w \models^+ \psi(x) \wedge \text{NEnt}_E^-(\varphi, \psi, w)). \end{aligned}$$

On the classical fragment these clauses recover Scott D2.

The positive T2 target is

$$T2^+ : \quad +G(x) \Rightarrow +G \text{Ess } x.$$

T2 is not automatic. Two explicit two-world S5 models refute $T2^+$ while satisfying the Gate-5 control stack together with strong A1 and R^+ . In the glut model, a property Z is both positively and negatively exemplified by one positive God-like witness but lacks positive support at another accessible God-like witness. In the gap model, $P(Z) = P(\neg Z) = \mathbf{N}$, while Z is positively exemplified locally but not necessarily shared. In both cases $A2^+$, $A3_T$, $G\text{-sup-v0.1}$, strong A1, and positive rigidity hold, yet $\text{NEnt}_E^+(G, Z)$ fails. Hence positive Godlikeness does not imply positive Godlikeness-as-essence merely by choosing a natural bilateral D2.

7.5 A sufficient recovery route for T2

For properties actually positively exemplified by positive God-like witnesses, define

$$\text{COMP}_P^G : \quad +G(x) \wedge +\psi(x) \Rightarrow (+P(\psi) \text{ or } -P(\psi)),$$

and

$$\text{CONS}_G^G : \quad +G(x) \wedge +\psi(x) \Rightarrow \text{not } -\psi(x).$$

Let REG_G be their conjunction.

Theorem 7.2 (Conditional recovery of positive T2). *Under $A1_L$, R^+ , $G\text{-sup-v}0.1$, and REG_G ,*

$$T2^+$$

holds.

Proof. Assume $+G(x)$ and let ψ be positively exemplified by x . Relevant positivity completeness gives $+P(\psi)$ or $-P(\psi)$. The latter alternative would, by $A1_L$, make $\neg\psi$ positively positive; Godlikeness would then yield $\neg\psi(x)$, contradicting relevant exemplification consistency. Hence $+P(\psi)$. Positive rigidity makes this support necessary, and every accessible positive God-like witness consequently positively exemplifies ψ . Thus $\text{NEnt}_E^+(G, \psi)$, as required by positive essence. $\square$

The glut and gap countermodels isolate the two regularity failures for this recovery route. I do not claim that REG_G is globally necessary for every model satisfying $T2^+$.

7.6 Bilateral necessary existence and A5

Define $NE\text{-sup-v}0.1$ by

$$\begin{aligned} w \models^+ NE(x) &\iff \forall \varphi (w \models^+ (\varphi \text{Ess } x) \Rightarrow w \models^+ \Box \exists^E y \varphi(y)), \\ w \models^- NE(x) &\iff \exists \varphi (w \models^+ (\varphi \text{Ess } x) \wedge w \models^- \Box \exists^E y \varphi(y)). \end{aligned}$$

These clauses recover Scott D3 classically.

For A5 distinguish

$$A5^+ : \quad +P(NE), \quad A5_T : \quad \text{Pos}_T(NE).$$

Unlike A3, the positive T3 branch needs only $A5^+$. Once $+G(x)$ is available, $G\text{-sup-v}0.1$ turns $+P(NE)$ directly into $+NE(x)$. Truth-only positivity of NE is unnecessary for that positive inference.

7.7 Positive necessary God-like existence

Theorem 7.3 (Positive T3 with symmetry). *Assume positive possible God-like existence, $T2^+$, $A5^+$, $G\text{-sup-v}0.1$, $NE\text{-sup-v}0.1$, and symmetry of the accessibility relation. Then*

$$T3^+ : \quad +\Box \exists^E x G(x).$$

No reflexivity or transitivity premise is required by this T3 route.

Proof. Fix a world w and an arbitrary successor z of w . Positive possible God-like existence supplies a world v accessible from w and an actual positive God-like witness x at v . By A5 and Godlikeness, x positively exemplifies NE ; by T2, G is a positive essence of x . Hence $NE\text{-sup-v0.1}$ supplies an actual positive G -instance at every successor of v .

By symmetry, wRv gives vRw , so one such actual positive G -instance exists already at w . The G -realization turns it into a positive God-like witness at w . Applying A5, T2, and $NE\text{-sup-v0.1}$ once more at w supplies an actual positive G -instance at the arbitrary successor z . Thus every successor of w has an actual positive God-like witness, establishing $T3^+$. $\square$

A finite two-world countermodel shows that reflexivity plus transitivity do not replace symmetry for this route. The frame $w_0Rw_0, w_0Rw_1, w_1Rw_1$ satisfies the encoded Scott premises of the reduced theorem other than symmetry, including $T2^+, A5^+, NE\text{-realization}, G\text{-realization}$, and possible Godlikeness, while $T3^+$ fails at w_0 .

On reflexive frames, $T3^+$ additionally supplies an actually existing positive God-like witness at every world, so the former interface GW is discharged. Reflexivity is therefore needed for this particular $T3^+ \Rightarrow GW$ step, not for the symmetry-only T3 theorem itself.

Combining the sufficient T2 route with Gate 5 gives the conditional positive Scott-control chain

$$A1_{\leftrightarrow} + A2^+ + A3_T + R^+ + REG_G + A5^+ + \text{Sym}(R) \Longrightarrow T3^+,$$

under the fixed higher-order semantics.

7.8 Essence-compressed modal collapse

Once $T2^+$ and $T3^+$ are available, the earlier reflection/rigidity proof can be compressed.

Theorem 7.4 (Essence-compressed collapse). *Under $T2^+, T3^+, G\text{-sup-v0.1}$, constant-property embedding, and reflexivity of the accessibility relation,*

$$MC^+$$

holds, and therefore MC^- holds by the Gate-4 collapse-equivalence theorem.

Proof. Assume $+\chi$ at w . Reflexivity together with T3 supplies an actual positive God-like witness x at w , and T2 makes G a positive essence of x . For the constant property $Z_\chi = \lambda y.\chi, +Z_\chi(x)$. Positive essence gives $NEnt_E^+(G, Z_\chi, w)$. T3 supplies a positive God-like witness at every accessible world; $G\text{-sup-v0.1}$ identifies each such witness with a positive G -instance. Necessary entailment therefore makes each such witness positively exemplify Z_χ , hence each accessible world positively supports χ . Therefore $+\Box\chi$. $\square$

This result refines the Gate-4 dependency map. The earlier REF^+ , $CONSG$, $COMP_P$, and R^+ assumptions are ingredients of one route for deriving T2 from lower-level principles; they are not separate premises of the essence-compressed collapse theorem after T2 and T3 have already been established.

7.9 Robustness and the new bottleneck

A5 illustrates genuine inconsistency tolerance. If $P(NE) = \mathbf{B}$, strong A1 also permits $P(\neg NE) = \mathbf{B}$. A positive God-like witness may therefore satisfy both $+NE(x)$ and $-NE(x)$, while the positive component remains sufficient for the T3 derivation.

Likewise, global positivity completeness and global consistency of arbitrary God-like exemplification are not semantic prerequisites for every model satisfying the positive T2/T3 chain. They are route-specific regularity assumptions used to recover T2 from the primitive lower-level axioms.

The decisive open question after Gate 6 is consequently not whether necessary existence can supply witnesses. It can, once T2 is available. The main remaining semantic-minimality problem is:

What are the weakest principled four-valued conditions under which $T2^+$ follows?

The full Gate-5 and Gate-6 specifications are maintained in `docs/GODLIKENESS_AND_POSSIBILITY.` and `docs/ESSENCE_AND_NECESSARY_EXISTENCE.md`.

8 Mechanization and Countermodels

STATUS: ESTABLISHED AT THE SEMANTIC LEVEL.

Gate 7 adds two complementary verification layers. I use a finite-model oracle for concrete countermodels and bounded exhaustive searches, and Lean for general semantic theorems that must not depend on a finite bound. Gate 8 extends the same discipline to comparative variants, modal-frame minimization, and the Fitting intension/extension reconstruction. This separation prevents the absence of a small countermodel from being mistaken for a proof and, conversely, allows an apparently successful proof interface to be tested for semantic vacuity.

8.1 Finite-model regression layer

The executable checker implements the four information values, involutive FDE negation, finite accessibility relations, actualist existence, four-valued property extensions and positivity, signed necessary entailment, support-based Godlikeness, positive essence, and the Gate-6 regularity conditions.

The regression suite machine-checks both two-world T2 countermodels from Gate 6. The glut model satisfies the encoded Gate-5 control stack

and $COMP_P^G$ but violates $CONS_G^G$, while the gap model satisfies the same control stack and $CONS_G^G$ but violates $COMP_P^G$. Both refute $T2^+$.

The checker also validates the Gate-5 obstruction to unrestricted positive T1. A one-world reflexive model with no actual entities and glutty positivity for a property and its complement satisfies strong A1 and $A2^+$, while positive support for the property does not yield positive possible actual exemplification. This is the executable counterpart of the distinction between mere positive support and truth-only positivity.

A first bounded search retains 204 models satisfying the original small control antecedent and verifies $T2^+$ in every retained model. A broader exhaustive two-world, one-entity G, Z family then varies positivity of Z and its complement independently while fixing FDE complement exemplification and full bilateral G -support semantics. In this broader family, 873 models satisfy

$$A1_L, \quad R^+, \quad COMP_P^G, \quad CONS_G^G,$$

and every one of them satisfies $T2^+$. Dropping any one of these four assumptions while retaining the other three produces a $T2^+$ countermodel within the same bounded family. I therefore treat this as bounded individual indispensability, not global model-theoretic minimality.

The same oracle now exports selected witnesses as directed Kripke graphs. Figure 1 shows the first countermodel isolated after dropping only $A1_L$: R^+ , $COMP_P^G$, and $CONS_G^G$ remain true while $T2^+$ fails. The two HTML-style node tables display the complete local FDE matrix. Each row records its value in $\{\mathbf{T}, \mathbf{F}, \mathbf{B}, \mathbf{N}\}$ together with the underlying positive and negative support bits. The four arrows are read directly from the model's complete two-world accessibility relation.

The checker also verifies the schema-level equivalence

$$MC^+ \leftrightarrow MC^-$$

on a negation-closed two-world test family. The equivalence concerns schemata closed under substitution by negation, not a pointwise identity for one isolated formula valuation.

Gate 8 adds comparative witnesses. The support/exact comparison confirms that both Gate-6 T2 countermodels are support-Godlike but fail project-internal exact positive Godlikeness, while a separate exact-Godlike fixture still permits genuine **B** gluts. A complete-S5 bilateral Anderson fixture has positive necessary Godlike existence while a contingent formula application refutes positive modal collapse. Separate Scott and Anderson S4-style two-world models are reflexive and transitive but not symmetric and refute the corresponding reduced T3 conclusions.

The Fitting branch adds two further finite witnesses. First, a three-world two-entity model uses a selected negation-closed extensional property domain and satisfies the admissible Fitting recovery stack. At the root

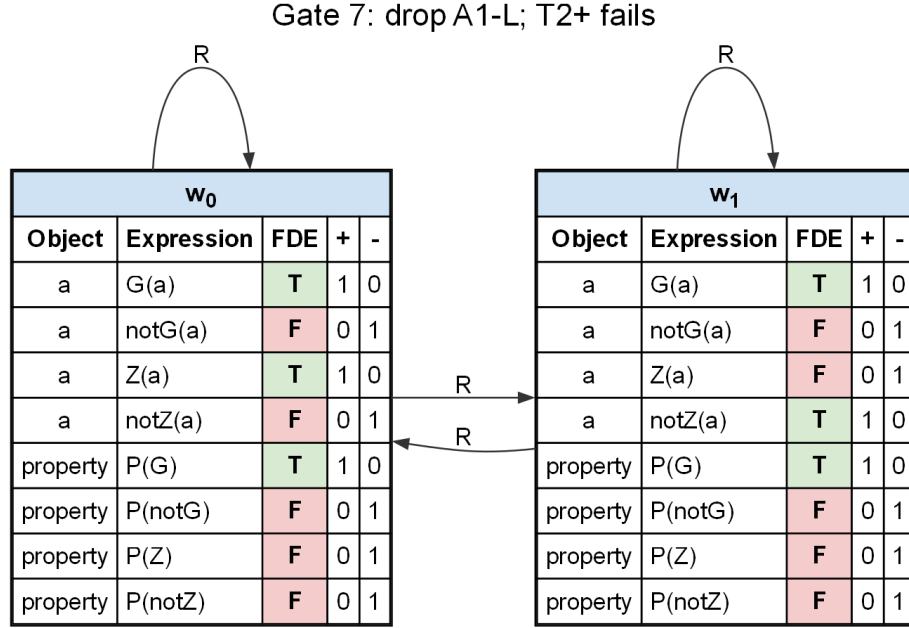


Figure 1: The first Gate-7 $T2^+$ countermodel found when only $A1_L$ is dropped. The graph is generated by `formal/finite/visualize_gate7_models.py` from the same matrix family used in the 873-model exhaustive regression.

world the current frozen G -extension is both possibly and necessarily actually exemplified de re, and de-dicto Godlikeness is possible, but de-dicto necessary Godlikeness fails because the extension of G changes at another accessible world. The model therefore separates the two readings when the explicit extension-stability condition introduced below is absent.

Second, a complete two-world S5 model satisfies the encoded admissible Fitting stack, extension stability, and positive necessary actual Godlikeness. Its selected extension domain nevertheless contains genuine inconsistent information: one admissible extension takes value **B** on an entity, together with its FDE negation. A separate contingent intension Q satisfies

$$+Q(a)@w_0$$

but not

$$+\Box Q(a)@w_0.$$

Thus the present admissible bilateral Fitting candidate has an explicit necessary-God / no-positive-collapse model. This is a finite countermodel to collapse for the candidate, not an unbounded theorem about every Fitting semantics.

8.2 Lean layer

The Lean package is pinned to Lean 4.30.0. It formalizes the four-valued carrier, FDE negation, conjunction and disjunction, bilateral relational modality, bilateral actualist quantification, signed necessary entailment, support-based Godlikeness, bilateral essence, and necessary-existence interfaces.

Lean proves the collapse-channel theorem generally at schema level and the Gate-5 truth-only possibility theorem

$$A1_R + A2^+ \implies T1_T.$$

The principal Gate-6 Scott recovery theorem is machine-checked without a finite bound:

$$NegExemplification + G\text{-sup} + A1_L + R^+ + REG_G \implies T2^+.$$

Gate 8 sharpens the later modal dependency. Once Scott $T2^+$ is available, Lean proves

$$\text{Sym}(R) + PossibleGod + T2^+ + A5^+ + NE\text{-sup} + G\text{-sup} \implies T3^+.$$

Reflexivity and transitivity are absent. Reflexivity enters only in the separate implication $T3^+ \Rightarrow GW$ and in the current essence-compressed collapse theorem

$$\text{Refl}(R) + T2^+ + T3^+ + G\text{-sup} + CONST \implies MC^+.$$

For project-internal exact positive Godlikeness, Lean proves

$$G\text{-exact}^+ \Rightarrow G\text{-sup}^+$$

and

$$G\text{-exact}^+ + R^+ \implies T2_{\text{exact}}^+.$$

For Anderson, the literature-grounded positive interfaces and project-specific bilateral negative evidence are mechanized separately. Under classical coherence the negative necessary-exemplification, Godlikeness, essence, necessary-actual-exemplification, and necessary-existence clauses recover the complements of their positive counterparts. The positive Anderson necessary-existence branch reduces to symmetry alone:

$$\begin{aligned} &\text{Sym}(R) + PossibleG_A + G_A\text{-realization} + P^+(G_A) + A2^+ \\ &\quad + NE_A\text{-realization} + P^+(NE_A) \implies T3_A^+. \end{aligned}$$

8.3 Fitting: from a type distinction to a domain obstruction

The Fitting comparison begins by distinguishing bilateral extensions from intensions:

$$Extension(Entity) \quad \text{and} \quad Intension(World, Entity) = World \rightarrow Extension(Entity).$$

The positivity predicate is typed over extensions, and *extensionAt* makes the passage from an intension to its extension at a world explicit. Signed necessary entailment, essence, and necessary existence are then reconstructed over rigid extensions, reflecting the extensional architecture of the mechanized Fitting variant [4].

A naive first lift quantified Fitting Godlikeness and the relevant consistency/completeness package over every inhabitant of the bilateral extension type. Lean exposes a problem with that choice. The unrestricted type contains the extension B satisfying

$$+B(x) \quad \text{and} \quad -B(x)$$

for every individual. Consequently Lean proves

$$CONS_{G,F}^{\text{all}} \implies \forall w \forall x \neg G_F^+(x, w),$$

and therefore also

$$REG_{G,F}^{\text{all}} \implies \forall w \forall x \neg G_F^+(x, w).$$

The earlier unrestricted implication from this regularity package to extensional essence remains a valid implication, but its intended Godlike antecedent is excluded by the package itself. I therefore do not count that unrestricted route as the substantive Fitting recovery result. The failed modeling choice is retained in the formal development because the obstruction is itself informative.

8.4 Admissible Fitting extensions

The corrected candidate uses a selected domain of admissible extensions, written $Adm(X)$, with FDE-negation closure

$$Adm(X) \implies Adm(\neg X).$$

This is a property-domain restriction rather than a global consistency postulate. In particular, admissible extensions may remain glutty, as the finite S5 witness above demonstrates.

Positive Fitting Godlikeness becomes

$$G_F^+(x, w) \quad \text{iff} \quad \forall X (Adm(X) \wedge +P(X)@w \Rightarrow +X(x)).$$

The relevant Fitting regularity package is likewise restricted to admissible extensions. Lean then proves the non-vacuous extensional essence theorem

$$\begin{aligned} & G\text{-admissible} + G_F\text{-realization} + A1_{L,F}^{adm} + REG_{G,F}^{adm} \\ \implies & (G_F^+(x, w) \Rightarrow Ess_F^+(ext_w G, x, w)). \end{aligned}$$

Unlike the current Scott-support T2 recovery theorem, no R^+ premise occurs. Once local positivity of an admissible rigid extension has been recovered, its membership relation does not itself vary across accessible worlds.

The bilateral negative evidence is also audited. Under classical coherence on the selected extension domain, Lean proves the recovery schemata

$$\begin{aligned} Ent_F^- & \leftrightarrow \neg Ent_F^+, \\ G_F^- & \leftrightarrow \neg G_F^+, \\ Ess_F^- & \leftrightarrow \neg Ess_F^+, \\ BoxExists_F^- & \leftrightarrow \neg BoxExists_F^+, \\ NE_F^- & \leftrightarrow \neg NE_F^+. \end{aligned}$$

These results validate the project-specific negative clauses at the classical boundary without imposing classicality on the four-valued candidate itself.

8.5 Fitting de re and de dicto

Let $ext_w G$ denote the extension of the Godlikeness intension frozen at the evaluation world. With a distinguished NE intension realizing admissible-domain necessary existence and positive A5 on its current extension, Lean proves

$$\begin{aligned} & \text{admissible Fitting stack} + \Diamond_{dere}^+ \exists^E x ext_w G(x) \\ \implies & \Box_{dere}^+ \exists^E x ext_w G(x). \end{aligned}$$

No reflexivity, symmetry, transitivity, seriality, or positivity-rigidity premise occurs. This is the project's four-valued counterpart of the K-level de-re behavior emphasized in the mechanized Fitting analysis [4].

The de-dicto claim is intentionally not identified with this theorem. It re-evaluates G at each accessible world. I isolate the required intension/extension bridge as

$$STAB_G : \quad wRz \Rightarrow ext_w G \equiv ext_z G.$$

Lean proves that $STAB_G$ converts de-dicto possibility to the frozen de-re premise and converts the de-re necessary conclusion back to de-dicto necessary Godlikeness. Hence

$$\begin{aligned} & \text{admissible Fitting stack} + STAB_G + \Diamond^+ \exists^E x G_F(x) \\ \implies & \Box^+ \exists^E x G_F(x), \end{aligned}$$

again without an S4 or S5 frame premise. The three-world finite witness shows that the stability condition is not merely decorative for this proof interface.

The minimized positive bridge admits a structural formulation. Lean defines $R_{F,adm}^+$ as forward positivity rigidity restricted to admissible rigid extensions and $R_{F,adm}^{+\leftarrow}$ as converse positivity transport along existing accessibility edges. It proves

$$\begin{aligned} R_{F,adm}^+ + G_F\text{-realization} &\implies Refl_G^+, \\ R_{F,adm}^{+\leftarrow} + G_F\text{-realization} &\implies Pers_G^+, \\ R_{F,adm}^+ + Sym(R) &\implies R_{F,adm}^{+\leftarrow}. \end{aligned}$$

Consequently $R_{F,adm}^+ + R_{F,adm}^{+\leftarrow}$ can replace primitive $STAB_G^+$ in the positive de-dicto theorem without any frame condition. The finite oracle checks this bridge exhaustively over 256 two-world, one-entity configurations in the classical negation-closed admissible fragment. Among 112 candidates satisfying both transports, 32 are non-symmetric; converse positivity transport is therefore strictly weaker than symmetry in the tested family.

8.6 Classical recovery and verification discipline

The Scott, Anderson, and admissible Fitting branches each have explicit recovery interfaces rather than relying on an informal assertion that the bilateral semantics “becomes classical.” Exact higher-order correspondence with the complete AFP/HOL developments is nevertheless a separate publication audit item.

The repository treats a countermodel or an obstruction theorem as a result rather than as a failed proof attempt. The unrestricted Fitting-comprehension obstruction is a particularly useful example: a theorem initially looked stronger because it lacked R^+ , but mechanization plus domain analysis revealed that the unrestricted regularity package excluded its intended witnesses. The admissible-domain reconstruction makes the subsequent theorem non-vacuous and is independently checked by finite models.

The explicit actualist quotient adds a third consistency check between the abstract and bounded layers. Lean’s `FittingActualistQuotient.lean` constructs quotient entities, proves bilateral extension push/pull round trips, characterizes existence factorization, and identifies quotient entailment with ordinary entailment on pullbacks. The matching finite fixture exhausts all sixteen bilateral extensions in a two-entity one-class model and compares all 256 ordered entailment pairs. It also checks that **B** and **N** survive quotienting and that split actual existence blocks representative-independent quotient interpretation.

The filter follow-up is split in the same way. The Lean quotient-filter module proves abstractly that point evaluation is a prime truth-order delta-

filter which designates **B** but is not complement-deciding because of **N**. The finite oracle enumerates all filter families on the sixteen-element two-point quotient algebra and separately checks the non-adjunctive two-filter condition and a non-vacuous *COMP*-failure fixture.

The deferred paired-neighborhood layer now has the same two-sided validation. Lean proves modal duality, exact relational representation, and complement-dual classical recovery for arbitrary world types. The finite regression exhausts all 256 local two-world pairs of neighborhood families and all 1024 relational operator comparisons. The complete Python regression suite currently contains 31 tests.

All executable and Lean sources are maintained under `formal/`, with CI configured to run both tracks. General claims are promoted to machine-checked status only after an unbounded Lean theorem is present, while finite witnesses are reported as finite witnesses with their model bounds explicit. Appendix A maps every claim of the form “Lean proves . . .” in this manuscript to the named declaration that discharges it, and every bounded count to the fixture that produces it. Where the text abbreviates a premise list as a “stack”, the Lean signature listed there is the authoritative statement.

8.7 Fitting assumption minimization

The admissible-domain reconstruction permits a second round of dependency analysis. The original recovery package

$$REG_{G,F}^{adm} = COMP_{P,F}^{G,adm} + CONS_{G,F}^{G,adm}$$

requires every admissible extension positively possessed by a positive Fitting-Godlike witness to be locally non-glutty. The proof of the extensional essence theorem, however, uses this consistency information only when positivity completeness classifies the possessed extension negatively. I therefore isolate the weaker condition

$$\begin{aligned} NegCons_{G,F}^{adm} : & \quad Adm(Y) \wedge G_F^+(x, w) \wedge +Y(x) \wedge -P(Y)@w \\ \implies & \quad \neg -Y(x). \end{aligned}$$

Writing

$$REG_{G,F}^{neg} := COMP_{P,F}^{G,adm} + NegCons_{G,F}^{adm},$$

Lean proves

$$CONS_{G,F}^{G,adm} \implies NegCons_{G,F}^{adm}$$

and the strengthened recovery theorem

$$\begin{aligned} & G\text{-admissible} + G_F\text{-realization} + A1_{L,F}^{adm} + REG_{G,F}^{neg} \\ \implies & G_F^+(x, w) \Rightarrow Ess_F^+(ext_w G, x, w). \end{aligned}$$

The corresponding de-re necessary-existence chain is machine-checked with the same targeted package. A one-world one-entity finite model separates the two consistency notions on the A1-L-only route: the sole admissible extension is genuinely glutty, the targeted negative-classification condition holds, full relevant exemplification consistency fails, and the realized positive Godlike extension is nevertheless an essence.

This weakening is itself directional. If the right A1 channel is restored, Lean proves

$$A1_{R,F}^{adm} + COMP_{P,F}^{G,adm} + NegCons_{G,F}^{adm} \implies CONS_{G,F}^{G,adm}.$$

Thus the reduction is not an unconditional weakening of a theory retaining both A1 directions. It instead exposes another division of labor between the two signed A1 flows: the A1-L recovery argument needs consistency only on the negative-classification branch, while A1-R supplies enough additional complement information to reconstruct the stronger consistency condition.

Why completeness cannot simply be renamed away. Inside this classification-based proof, $COMP_{P,F}^{G,adm}$ supplies exactly the gap branch: a relevant extension either already has positive positivity support or is negatively classified so that A1-L can activate the complement route. Replacing the disjunction by a condition of the form

$$\neg +P(Y)@w \implies -P(Y)@w$$

would therefore not constitute a genuine weakening at the classical meta-level; on the same relevant domain it is equivalent to the branch coverage already expressed by completeness. A real reduction of $COMP$ must consequently change the recovery mechanism itself.

An alternative recovery route. I introduce a second, deliberately non-classificatory interface. Let admissible-property indiscernibility among positive Fitting-Godlike individuals be

$$\begin{aligned} Ind_G^{adm} : & \quad Adm(Y) \wedge G_F^+(x, w) \wedge G_F^+(y, w) \\ & \implies (+Y(x) \leftrightarrow +Y(y)). \end{aligned}$$

This condition does not assign positive or negative positivity status to Y . Instead, it constrains the selected extensional property domain: admissible rigid extensions cannot positively distinguish two positive Godlike individuals at the same source world.

Lean proves

$$G\text{-admissible} + G_F\text{-realization} + Ind_G^{adm} \implies G_F^+(x, w) \Rightarrow Ess_F^+(ext_w G, x, w).$$

No A1 direction, no positivity completeness, and no exemplification-consistency premise occurs. The proof is extensional. If Y contains the selected Godlike individual x , any entity y in the frozen current G -extension is also positive Godlike at the source world by G -realization. Indiscernibility then transfers positive Y -membership from x to y , yielding exactly the rigid necessary entailment required by Fitting essence.

The same route continues through necessary existence. Lean machine-checks a frame-free de-re possibility-to-necessity theorem from G -admissibility, G -realization, Ind_G^{adm} , NE-admissibility, NE-realization, and positive A5, with no $COMP$, A1, or consistency premise. Positive G -extension persistence and reflection then provide the same de-dicto bridge as before.

Positive-profile quotient structure. The indiscernibility premise can itself be derived from a more structural condition on the selected property domain. Define positive-profile equivalence at a world by

$$x \approx_w^+ y$$

iff every positively supported admissible extension contains x and y alike. Lean proves that $\approx_w^+$ is reflexive, symmetric, and transitive. Positive Fitting-Godlike individuals automatically lie in the same $\approx_w^+$ -class, because each of them positively exemplifies every positively supported admissible extension.

I then impose the quotient-style saturation condition

$$\begin{aligned} Sat_P^{adm} : \quad & Adm(Y) \wedge x \approx_w^+ y \\ & \implies (+Y(x) \leftrightarrow +Y(y)). \end{aligned}$$

Lean proves

$$Sat_P^{adm} \implies Ind_G^{adm},$$

and therefore the entire alternative essence/NE route follows from this property-domain condition. This gives the previously primitive indiscernibility interface a quotient-semantical interpretation rather than treating it as an isolated recovery axiom.

The quotient is in fact bilateral. Since the substantive admissible semantics is already closed under FDE negation, applying Sat_P^{adm} to $\neg Y$ transfers constancy of positive membership into constancy of negative membership of Y . Lean therefore proves

$$\boxed{Sat_P^{adm} \iff Sat_{\pm}^{adm}},$$

where $Sat_{\pm}^{adm}$ requires both support channels of every admissible extension to be constant on the same positive-profile classes. Thus each admissible bilateral extension genuinely factors through one entity quotient rather than merely sharing a positive projection.

A one-world, three-entity finite fixture makes the result nontrivial. Two distinct entities a and b are both positive Godlike and form one non-singleton positive-profile class, while c is not Godlike. The negation-closed admissible domain consists of the rigid extensions

$$\emptyset, \quad \{a, b\}, \quad \{c\}, \quad \{a, b, c\}.$$

Every admissible extension respects the $\{a, b\}$ profile class, so Sat_P^{adm} , its bilateral counterpart, and Ind_G^{adm} hold. Nevertheless the universal extension is a genuine positivity gap:

$$\neg +P(\{a, b, c\}) \quad \text{and} \quad \neg -P(\{a, b, c\}),$$

so $COMP_{P,F}^{G,adm}$ fails at an extension containing both Godlike individuals. The current G -extension $\{a, b\}$ is still an essence of each, and the finite NE/A5 interface remains satisfied. Hence the profile-saturation route bypasses positivity completeness non-vacuously even when Godlike indiscernibility has real two-individual content.

Algebraic stability of the quotient. The quotient interpretation is compatible with the basic FDE property operations. At the extension level define conjunction and disjunction pointwise by

$$\begin{aligned} +(X \wedge_F Y)(x) &\Leftrightarrow +X(x) \wedge +Y(x), & -(X \wedge_F Y)(x) &\Leftrightarrow -X(x) \vee -Y(x), \\ +(X \vee_F Y)(x) &\Leftrightarrow +X(x) \vee +Y(x), & -(X \vee_F Y)(x) &\Leftrightarrow -X(x) \wedge -Y(x). \end{aligned}$$

Lean proves that, for each fixed positive-profile quotient, the bilateral extensions that factor through it contain the FDE top and bottom extensions and are closed under FDE negation, conjunction, and disjunction. In this precise preservation sense the quotient-respecting extensions form a De-Morgan-style subalgebra of the full bilateral extension algebra.

This theorem does not itself impose algebraic closure on Adm . It establishes the weaker and methodologically prior fact that adding FDE compounds of quotient-respecting extensions cannot break the quotient factorization. Whether the intended Fitting property domain should actually be closed under all of these operations remains a separate semantic choice.

Relation to Fitting's ultrafilter structure. The published mechanized analysis proves that Fitting's extensional positivity predicate constitutes a δ -ultrafilter [4]. Its classical maximality component chooses an extension or its complement as positive. To compare that architecture with the bilateral reconstruction, I isolate a four-valued complement-decision interface:

$$Adm(Y) \Longrightarrow (+P(Y) \vee +P(\neg Y)).$$

This condition alone is not yet a four-valued ultrafilter definition. More importantly, the bilateral setting changes what the complement branch

supplies: positive support for $\neg Y$ yields negative support information about membership in Y , but it does not by itself erase positive support for Y .

Lean shows that complement decision combined with full relevant exemplification consistency already recovers local positive reflection at Godlike witnesses, and hence $COMP_{P,F}^{G,adm}$. Independently, complement decision together with $A1_R$ also yields $COMP_{P,F}^{G,adm}$. Thus the ultrafilter-style route explains why completeness is natural at the classical Fitting boundary, but it does not account for the genuinely classification-free route above. The quotient-saturation condition and ultrafilter maximality therefore locate two different ways of controlling the property domain.

A separate minimization concerns de-dicto stability. Full bilateral extension stability

$$STAB_G^\pm : \quad wRz \Rightarrow ext_w G \equiv ext_z G$$

contains more information than the positive theorem uses. I split the positive channel into persistence and reflection:

$$\begin{aligned} Pers_G^+ & : \quad wRz \wedge +ext_w G(x) \Rightarrow +ext_z G(x), \\ Refl_G^+ & : \quad wRz \wedge +ext_z G(x) \Rightarrow +ext_w G(x). \end{aligned}$$

Their conjunction is written $STAB_G^+$. Lean proves that bilateral stability implies $STAB_G^+$, but the two positive directions are exactly what the de-dicto proof consumes: reflection converts de-dicto possibility into frozen de-re possibility, whereas persistence converts frozen de-re necessity back into de-dicto necessity. Consequently the minimized classification-route theorem is

$$\begin{aligned} & G\text{-admissible} + G_F\text{-realization} + A1_{L,F}^{adm} + REG_{G,F}^{neg} \\ & + NE_F\text{-admissible} + NE_F\text{-realization} + A5_F^+ + STAB_G^+ + \Diamond^+ \exists^E x G_F(x) \\ & \implies \Box^+ \exists^E x G_F(x). \end{aligned}$$

No negative-extension stability and no S4/S5 frame condition occurs. Replacing the classification block by Ind_G^{adm} , or more structurally by Sat_P^{adm} , gives an alternative theorem with the same positive stability interface.

The positive stability interface can now be derived from a more structural premise. Let $R_{F,adm}^+$ restrict positive-property rigidity to the selected admissible extension domain:

$$Adm(X) \wedge +P_w(X) \wedge wRz \implies +P_z(X).$$

Lean proves the asymmetric pair

$$\begin{aligned} G_F\text{-realization} + R_{F,adm}^+ & \implies Refl_G^+, \\ G_F\text{-realization} + R_{F,adm}^+ + Sym(R) & \implies Pers_G^+. \end{aligned}$$

Forward rigidity is enough for reflection because every extension positive at the source remains positive at the target, where target Godlikeness consumes

it. Persistence requires only the converse positivity transport, not the converse edge itself. Write $R_{F,adm}^{+\leftarrow}$ for

$$Adm(X) \wedge wRz \wedge +P_z(X) \implies +P_w(X).$$

Lean proves

$$\begin{aligned} R_{F,adm}^+ + Sym(R) &\implies R_{F,adm}^{+\leftarrow}, \\ G_F\text{-realization} + R_{F,adm}^{+\leftarrow} &\implies Pers_G^+, \\ G_F\text{-realization} + R_{F,adm}^+ + R_{F,adm}^{+\leftarrow} &\implies STAB_G^+. \end{aligned}$$

Substituting the last implication into the minimized theorem yields positive de-dicto necessity without a frame premise. The older $R_{F,adm}^+ + Sym(R)$ package remains sufficient because it entails the converse transport, but frame symmetry is not part of the sharper proof interface.

The weakening of stability is strict even under a stronger surrounding control theory. A complete two-world S5 fixture satisfies both A1 directions and the original full admissible regularity package. The positive membership set of G is $\{a\}$ at both worlds, but negative information about a non-Godlike individual changes from falsity support to a gap. Hence $STAB_G^+$ holds while $STAB_G^\pm$ fails, and positive necessary de-dicto Godlikeness still holds. This finite witness shows that the positive theorem is genuinely insensitive to drift confined to the negative extension channel.

A second finite regression independently checks the derived bridge over all 256 two-world, one-entity configurations on the negation-closed classical admissible domain $\{\mathbf{T}, \mathbf{F}\}$. Of these, 152 satisfy $R_{F,adm}^+$, and every one satisfies $RefI_G^+$; 152 satisfy $R_{F,adm}^{+\leftarrow}$, and every one satisfies $Pers_G^+$. There are 112 candidates satisfying both transports. Only 80 are symmetric, leaving 32 asymmetric candidates with complete positive G -stability. An explicit one-way frame with non-vacuous positivity witnesses the strict bounded separation from symmetry. A second fixture satisfies converse transport without forward transport, preserving persistence while reflection fails. These are bounded separation results, not global necessity claims.

These results do not establish global minimality. They reveal two distinct recovery architectures inside the same Fitting candidate: one resolves positivity gaps by classification, while the other factors the admissible bilateral property domain through positive-property profiles and thereby makes the gap irrelevant to extensional essence.

8.8 Fitting quotient-domain algebra

The positive-profile saturation condition introduced above has a stronger bilateral consequence. The selected admissible domain is already closed under FDE negation. Hence, if an admissible extension Y is positively constant on a positive-profile equivalence class, then the admissible complement $\neg Y$ is

positively constant on that class as well. Since positive membership in $\neg Y$ is negative membership in Y , Lean proves

$$Sat_P^{adm} \iff Sat_{\pm}^{adm},$$

where $Sat_{\pm}^{adm}$ requires both positive and negative membership of every admissible bilateral extension to be constant on the same positive-profile classes. Thus the quotient interpretation is genuinely bilateral rather than merely a statement about the positive projection of an extension.

The quotient is also stable under the basic FDE extension operations. Define pointwise

$$\begin{aligned} +(X \wedge_F Y)(x) &\iff +X(x) \wedge +Y(x), & -(X \wedge_F Y)(x) &\iff -X(x) \vee -Y(x), \\ +(X \vee_F Y)(x) &\iff +X(x) \vee +Y(x), & -(X \vee_F Y)(x) &\iff -X(x) \wedge -Y(x). \end{aligned}$$

Lean proves that, for each fixed positive-profile quotient, the bilateral extensions that factor through it contain the FDE top and bottom extensions and are closed under FDE negation, conjunction, and disjunction. In this preservation sense they form a De-Morgan-style subalgebra of the full bilateral extension algebra.

This result does not derive profile saturation from ordinary algebraic closure. A finite countermodel makes the distinction explicit. Let the selected domain over two entities be the complete classical four-extension algebra

$$\emptyset, \quad \{a\}, \quad \{b\}, \quad \{a, b\}.$$

It contains top and bottom and is closed under FDE negation, conjunction, and disjunction. Let positivity support only the universal extension $\{a, b\}$. Then a and b have the same positive-property profile, but the admissible singleton $\{a\}$ distinguishes them. Therefore profile saturation fails although the selected extension domain is fully closed under the basic FDE algebra.

Consequently the verified dependency is asymmetric:

$$\boxed{Sat_P^{adm} \implies \text{algebraically stable quotient factorization,}}$$

while

$$\boxed{\text{FDE algebra closure} \not\Rightarrow Sat_P^{adm}.}$$

The quotient route therefore still requires an independent extensionality or saturation principle. Algebraic closure is compatible with that principle and preserves it, but does not explain its origin.

This separation also sharpens the comparison with the classical Fitting ultrafilter result [4]. Ultrafilter-style maximality constrains which extensions count as positive; profile saturation instead constrains how the selected extension domain may distinguish entities. The two mechanisms operate on different semantic interfaces, and neither should be silently identified with the other in the four-valued reconstruction.

8.9 Entailment and the profile quotient

The preceding quotient construction raises a natural question: can the profile-saturation condition itself be generated by closing the admissible property domain under Fitting-style extensional entailment? The classical comparison suggests caution. In the mechanized Fitting analysis, the δ -filter is a family of positive extensions: it contains the universal extension, excludes the empty extension, is upward closed under supersets, and is closed under intersections [4]. This is closure of the positivity filter, not closure of the ambient property domain. The present $A2_F^+$ interface already captures the corresponding monotonicity of positive support under extensional entailment.

I therefore distinguish that source-grounded filter behavior from the stronger domain principle

$$\begin{aligned} EntCl^{adm} : \quad & Adm(X) \wedge \forall w \, Entail_F^+(w, X, Y) \\ & \implies Adm(Y). \end{aligned}$$

The stronger principle is unsuitable once FDE bottom is admitted. Since $\perp_F$ has no positive members,

$$Entail_F^+(w, \perp_F, Y)$$

holds vacuously for every world w and every rigid extension Y . Lean consequently proves

$$\boxed{EntCl^{adm} + Adm(\perp_F) \implies \forall Y \, Adm(Y).}$$

Thus naive entailment closure of the admissible property universe reconstructs unrestricted bilateral comprehension. Combined with relevant exemplification consistency, the earlier comprehension obstruction returns immediately:

$$\boxed{EntCl^{adm} + Adm(\perp_F) + CONS_{G,F}^{G,adm} \implies \forall w, x \, \neg G_F^+(x, w).}$$

The distinction between filter closure and domain closure is therefore substantive rather than terminological.

A finite separation strengthens this negative result. Over two entities I take the complete $4^2 = 16$ bilateral extension universe as admissible. Both entities actually exist at one reflexive world, and positive support is assigned exactly to extensions whose positive channel contains both entities. This model satisfies $A2_F^+$, full domain-level entailment closure, closure under the basic FDE extension algebra, and quotient-compatible actual existence. Positivity nevertheless gives the two entities the same positive-property profile while an admissible singleton extension distinguishes them. Hence

$$\boxed{A2_F^+ + EntCl^{adm} + \text{FDE algebra closure} + \text{profile-compatible existence} \not\Rightarrow Sat_P^{adm}}$$

in this explicit finite model. Entailment closure therefore does not generate the missing quotient structure.

The positive result takes a different form. For each world w , define a bilateral profile closure on arbitrary rigid extensions by

$$\begin{aligned} +Sat_w(Y)(x) &\iff \exists y (x \approx_w^+ y \wedge +Y(y)), \\ -Sat_w(Y)(x) &\iff \exists y (x \approx_w^+ y \wedge -Y(y)). \end{aligned}$$

Lean proves that Sat_w is extensive and monotone and is idempotent up to bilateral extensional equivalence. Every $Sat_w(Y)$ respects the positive-profile quotient. More strongly, if Z already respects the quotient and Y is support-included in Z , then

$$Sat_w(Y) \leq_k Z,$$

where $\leq_k$ is the FDE information order

$$X \leq_k Y \iff X^+ \subseteq Y^+ \wedge X^- \subseteq Y^-,$$

in which both support channels grow together. Thus $Sat_w(Y)$ is the least quotient-respecting bilateral extension above Y in the information order.

This yields an exact fixed-point characterization of the earlier domain condition:

$$\boxed{Sat_P^{adm} \iff \forall w, Y (Adm(Y) \rightarrow Sat_w(Y) \equiv Y).}$$

The quotient route can therefore be read as selecting admissible properties from the fixed points of a canonical closure operator induced by positive-property observational equivalence.

Actualist quantification introduces one further boundary. Fitting entailment quantifies only over entities that actually exist at accessible worlds. Even if two entities lie in the same positive-profile class, the existence predicate can still distinguish them. I isolate the compatibility condition

$$ExSat_P : \quad wRz \wedge x \approx_w^+ y \implies (E(z, x) \leftrightarrow E(z, y)).$$

Under this premise Lean proves

$$\boxed{Entail_F^+(w, X, Y) \implies Entail_F^+(w, Sat_w(X), Sat_w(Y)).}$$

Hence actualist Fitting entailment descends through the profile closure when actual existence itself factors through the same quotient.

A second finite fixture shows why the existence premise matters. Two entities have the same positive-property profile, but at the sole accessible target world only one actually exists. An extension whose only positive member is the non-actual representative then entails an empty-positive

extension vacuously. Profile closure transfers that positive membership to the equivalent actual representative, after which the saturated entailment fails. Thus the existence-compatibility condition is not merely a notational convenience, although the finite witness does not establish its global necessity for every alternative quotient construction.

The quotient can now be made literal rather than represented only by fixed points. Lean defines

$$Q_w = E / \approx_w^+$$

as a quotient type and the canonical projection $x \mapsto [x]_w$. Every bilateral extension on Q_w pulls back to a profile-respecting extension on entities. Conversely, every profile-respecting bilateral extension pushes forward to Q_w , with representative-independence proved separately for its positive and negative support channels. Both push-pull composites recover their input up to bilateral extensional equivalence. In particular, every profile closure $Sat_w(Y)$ has an explicit quotient representation.

The existence premise also receives an exact characterization. Write $ExFact(w, z)$ when there is a predicate $E_{w,z}^Q$ on Q_w such that

$$E_{w,z}^Q([x]_w) \leftrightarrow E(z, x).$$

Lean proves

$$\boxed{ExSat_P \iff \forall w, z (wRz \rightarrow ExFact(w, z)).}$$

Thus profile-compatible actual existence is exactly what makes actualist quantification over quotient entities independent of representatives, rather than an unexplained auxiliary closure clause.

Finally, define $Entail_{F,Q}^+$ directly on quotient extensions, quantifying over accessible worlds and classes satisfying $E_{w,z}^Q$. Lean proves

$$\boxed{Entail_{F,Q}^+(w, Q_X, Q_Y) \iff Entail_F^+(w, \pi_w^* Q_X, \pi_w^* Q_Y).}$$

For profile-respecting entity extensions, pushforward gives the converse representation as well. The earlier one-way closure-descent theorem is therefore subsumed by a two-way entailment equivalence on the quotient-respecting fragment.

The finite regression realizes one non-singleton profile class. It checks all sixteen saturated bilateral extensions and all 256 ordered entailment pairs, while both **B** and **N** remain possible quotient values. When actual existence splits the class, no representative-independent quotient existence predicate can be formed and the earlier entailment failure reappears.

The resulting architecture is more precise than the initial entailment-closure hypothesis. Positivity-filter closure remains governed by $A2_F^+$; the admissible property universe should not be closed indiscriminately under

entailment; profile saturation is instead a fixed-point condition for a canonical closure operator; and actualist entailment is represented directly on the quotient exactly when actual existence factors through the relevant profile classes. The next open problem is the algebraic/filter structure of these quotient fixed points.

8.9.1 A four-valued delta-filter on quotient extensions

The explicit quotient makes the filter question precise. A lattice filter does not belong to the information order $\leq_k$ just used for the profile hull, but to the FDE truth order

$$X \leq_t Y \iff X^+ \subseteq Y^+ \wedge Y^- \subseteq X^-,$$

in which positive support grows upwards while negative support shrinks. The two orders must not be conflated: Sat_w is a closure operator for $\leq_k$, whereas the filter structure is $\leq_t$ -ordered.

A positive delta-filter is proper, upward closed in $\leq_t$, and closed under FDE conjunction. Primeness adds

$$X \vee_F Y \in F \implies X \in F \text{ or } Y \in F.$$

For classically coherent extensions, where negative membership is exactly absent positive membership, Lean proves that $\leq_t$ reduces to ordinary positive-extension inclusion. Thus this is a genuine four-valued lift of the classical filter order rather than a replacement unrelated to it.

The crucial separation is already general. For any quotient class q , let

$$F_q(X) \iff +X(q).$$

Lean proves that F_q is a proper prime delta-filter. It designates the constant glut extension **B**. The constant gap extension **N** is fixed by FDE negation and neither it nor its negation is designated. Hence

$$\boxed{\text{prime delta-filter} \not\Rightarrow \forall X (X \in F \vee \neg_F X \in F)}.$$

Every inhabited positive-profile quotient therefore carries a prime filter that does not return to the complement-decision route isolated earlier.

The complete two-class quotient algebra provides a bounded structural audit. Among all families of its sixteen bilateral extensions, there are 166 proper truth-order upsets, 15 ordinary delta-filters, four prime delta-filters, 112 proper non-adjunctive two-filters, and ten prime two-filters. All four prime ordinary filters fail complement decision. The union of the two point-evaluation filters is a prime two-filter but not an ordinary filter. Conversely, the ordinary filter requiring positive membership at both classes has two non-vacuous Godlike points, while a $(\mathbf{T}, \mathbf{N})$ extension is possessed at the

first point but neither it nor its FDE negation belongs to the filter. Thus local *COMP* fails in that fixture.

The filter question is consequently narrowed rather than closed: four-valued properness, conjunction closure, and even primeness preserve the decisive gap. This settles the question for the abstract quotient algebra, where the filter is a family of quotient extensions carrying no positivity predicate of its own. The next subsection asks the corresponding question one level down, where the filter is the positivity predicate itself on the selected admissible domain, and where maximality can therefore interact with the Godlikeness and consistency interfaces.

8.10 A four-valued δ -filter on the fixed-point algebra

Section 8.9.1 placed a four-valued δ -filter on the abstract quotient algebra, where the filter is an arbitrary family of quotient extensions. The closure construction also supports the corresponding question one level down, on the selected admissible domain itself, where the filter is the positivity predicate P rather than an auxiliary family. This is the level at which Fitting's extensional δ -filter lives, and the level at which maximality can interact with Godlikeness. Stating it here also keeps positivity-filter closure separate from closure of the ambient property domain, the distinction forced by the entailment result above.

The order is the FDE truth order $\leq_t$ of Section 8.9.1, not the information order $\leq_k$ used by the profile hull. Lean verifies that the pointwise FDE operations introduced above are meet and join for $\leq_t$, while FDE negation reverses this order. Thus Sat_w is a closure operator in the information order, whereas the filter structure on its fixed points is truth-ordered.

I call the selected domain an admissible FDE algebra when it contains $\top_F, \perp_F$ and is closed under FDE conjunction and disjunction; negation closure is already built into the admissible semantics. A profile δ -filter then combines this algebra condition with profile saturation and, at every world, the laws

$$\begin{aligned} &+P(\top_F), \quad \neg +P(\perp_F), \\ &+P(X) \wedge X \leq_t Y \implies +P(Y), \end{aligned}$$

and

$$+P(X) \wedge +P(Y) \implies +P(X \wedge_F Y),$$

for admissible X, Y . The preceding fixed-point theorem then places every carrier extension among the fixed points of Sat_w .

This definition has the intended classical recovery. Let $X \leq_+ Y$ abbreviate $X^+ \subseteq Y^+$, the ordinary subset order used by the classical δ -filter [4]. Truth-order inclusion always entails $\leq_+$. If admissible extensions are classically coherent, positive inclusion conversely entails truth-order inclusion because negative membership is complementary to positive membership.

Lean therefore proves equivalence between the FDE truth-order filter and the source-style positive-set filter on the coherent fragment.

The bilateral extension adds the polarity equation

$$-P(X) \iff +P(\neg_F X).$$

Lean derives both admissible A1 directions and shows that negative support forms the dual ideal: it contains $\perp_F$, excludes $\top_F$, is downward closed in $\leq_t$, and is closed under FDE disjunction. This equation still permits both positivity gluts and positivity gaps; it is not a bivalence condition.

Ultrafilter maximality remains separate:

$$\forall X \in \text{Adm}, \quad +P(X) \vee +P(\neg_F X).$$

As this is precisely complement decision, the earlier dependency result applies. Lean proves

$$\boxed{\text{profile } \delta\text{-ultrafilter} + \text{CONS}_{G,F}^{G,adm} \implies \text{COMP}_P^{G,adm}},$$

and obtains the same conclusion with $A1_R$ in place of relevant exemplification consistency. The proof consumes only the maximality clause; the filter and saturation conditions are carried by the definition but not used. Hence maximality returns to the classification route on its own, and it is not part of the genuinely *COMP*-independent filter core.

A finite witness separates these notions. Its admissible algebra is the 12-element product

$$\{\mathbf{T}, \mathbf{F}, \mathbf{N}\} \times \{\mathbf{T}, \mathbf{F}, \mathbf{N}, \mathbf{B}\}$$

over two entities. It is closed under the FDE operations and contains both gap and glut information. Positive support is the truth-order filter generated by $(\mathbf{T}, \mathbf{N})$. This generator distinguishes the two entities, so the induced positive-profile quotient is discrete and all admissible extensions are fixed points. The model has a positive Godlike witness and satisfies relevant exemplification consistency. Nevertheless, for the split extension $A = (\mathbf{T}, \mathbf{F})$,

$$+A(a), \quad \neg +P(A), \quad \neg -P(A).$$

Therefore the profile δ -filter plus relevant consistency does not imply $\text{COMP}_P^{G,adm}$, while the corresponding ultrafilter does so generally. This strict separation is the main reason to keep filter closure, profile closure, and maximality as three distinct interfaces.

9 Related Work

The project sits at the intersection of two mature lines of work whose combination motivates the present question. First, Gödel–Scott and related

ontological theories have been extensively reconstructed and analyzed with computational tools, including comparisons of Scott, Anderson, and Fitting variants and their behavior with respect to modal collapse [5, 4]. Recent machine-checked analysis further separates positivity rigidity from ultrafilter structure as explanations of collapse [3].

The Anderson/Fitting comparison is especially relevant to Gate 8. Anderson’s original emendation and Fitting’s intensional treatment provide the historical controls [1, 7]. The computer-supported analysis of Benzmüller and Fuenmayor and its current AFP development distinguish intensional properties from their extensions and use this distinction to compare the two non-collapsing variants [4, 9]. In the source IHOML treatment, Scott and Anderson apply positivity to intensional properties, whereas Fitting moves positivity to property extensions and distinguishes de-re from de-dicto formulations. The present project follows that typing discipline rather than modeling Fitting by merely changing axioms over the existing Scott property type.

The four-valued reconstruction introduces an additional issue that is absent from a purely Boolean extension space. If the Fitting property domain is identified with the unrestricted type of all bilateral extensions, that type contains arbitrary glut constructions. In particular, a universal glut extension makes the project’s relevant exemplification-consistency condition incompatible with any positive Godlike witness. The admissible-domain reconstruction in Section 8 should therefore be read as a semantic-domain choice forced by the interaction between Fitting’s extensional architecture and unrestricted bilateral comprehension, not as an attribution to Fitting of an historical four-valued restriction. The selected domain remains closed under FDE negation and may contain genuinely glutty extensions.

This also sharpens the de-re/de-dicto comparison. The project reproduces a K-style, frame-free de-re possibility-to-necessity route at the bilateral extensional interface, while the de-dicto route requires an explicit stability bridge connecting the extension of the Godlikeness intension across accessible worlds. The source literature already warns against conflating neighboring de-re and de-dicto formulations [4]; the present finite separation model makes that distinction executable in the four-valued setting.

The same literature motivates a methodological distinction in the Anderson branch. Anderson strengthens Godlikeness by classifying the necessary properties of an individual through positivity and modifies essence accordingly, rather than merely adding a stronger Scott-style reflection principle [4]. The bilateral negative clauses used here are consequently a project-specific FDE reconstruction around a literature-grounded positive interface; they are not attributed to Anderson himself.

Second, Belnap–Dunn/FDE semantics provides the canonical four-valued distinction between truth-only, falsity-only, both, and neither [2, 6]. Font supplies a foundational De Morgan-lattice analysis, while recent work on up-

sets and (n)-filters gives a broader algebraic comparison class for four-valued consequence [8, 13]. These sources motivate the next filter experiment but do not by themselves identify logical consequence filters with the present higher-order positivity predicate. Modal extensions with Belnapian values have been developed using Kripke and algebraic semantics [11, 14]. Particularly close in semantic neighborhood is Petrukhin’s recent treatment of essence and accident modalities in an S5-based four-valued setting [12].

The actualist layer also has an independent source boundary. The AFP control uses existence relativization to restrict individual quantifiers at each world [9]. Work on partial semantics for varying-domain quantified modal logic shows why non-existence creates a genuine semantic choice rather than a harmless notational variant [10]. Neither source is claimed to contain the profile-quotient factorization theorem proved here.

Novelty status. A first reproducible prior-art pass now covers exact four-valued/Gödel queries, the primary Anderson and Fitting sources, the current AFP control development, De Morgan filter literature, and actualist varying-domain semantics. It did not locate a direct predecessor combining all of these components, but a negative search result does not establish absence. The manuscript therefore states structural and mechanization results only. In particular, the symmetry-only T3 reductions, the bilateral Anderson candidate, the unrestricted Fitting-comprehension obstruction, the admissible-domain repair, and the profile quotient remain project results without a global novelty claim.

10 Conclusion and Research Agenda

This manuscript treats the Gödel–Scott theory not merely as an object to be translated from two values to four, but as a controlled test case for locating informational dependencies that classical bivalence suppresses.

By version 0.19 the Scott reconstruction and verification branches reach positive necessary Godlike existence, the associated collapse analysis, a mechanically supported comparison of three different Godlikeness/essence architectures, and an exact paired-neighborhood representation of the relational modal control. Gates 1–4 establish the bilateral propositional and modal semantics, decompose A1 and A4, falsify the original collapse-separation conjecture under involutive FDE negation, and isolate glut/gap failures in the classical reflection step. Gate 5 reconstructs truth-only positive-property exemplification, a control reading of A3, possible positive Godlike existence, and support-based bilateral Godlikeness.

Gate 6 identifies the sharper Scott bottleneck. A natural bilateral definition of essence recovers Scott D2 classically, but it does not make Godlikeness an essence automatically. Two explicit two-world S5 models satisfy the Gate-

5 control conditions together with strong A1 and positive rigidity while refuting

$$T2^+ : \quad +G(x) \Rightarrow +G \text{ Ess } x.$$

The glut and gap models fail for different informational reasons. A targeted regularity package recovers T2, and Gate 7 changes the epistemic status of this result by machine-checking

$$\text{NegExemplification} + G\text{-sup} + A1_L + R^+ + REG_G \Longrightarrow T2^+.$$

Lean also verifies the earlier truth-only possibility theorem

$$A1_R + A2^+ \Longrightarrow T1_T.$$

Thus the division of labor inside Scott's A1 is represented in independent formal derivations.

Once T2 is available, the remaining Scott necessary-existence branch requires less frame structure than the original S5 presentation suggested. Gate 8 proves

$$\begin{aligned} & \text{Sym}(R) + \text{PossibleGod} + T2^+ + A5^+ + NE\text{-sup} + G\text{-sup} \\ & \Longrightarrow T3^+. \end{aligned}$$

Neither reflexivity nor transitivity is a premise. A reflexive-transitive but non-symmetric finite model refutes the corresponding S4 replacement. Reflexivity reappears later, where T3 supplies the global witness condition and where the current essence-compressed positive collapse theorem uses

$$\text{Refl}(R) + T2^+ + T3^+ + G\text{-sup} + CONST \Longrightarrow MC^+.$$

The informative negative collapse schema then follows from the independently machine-checked collapse-channel equivalence.

The finite-model layer provides complementary negative evidence. The broader exhaustive two-world, one-entity G, Z family retains 873 models satisfying the full current Scott T2 recovery package, and every retained model satisfies $T2^+$. Dropping any one of $A1_L$, positive rigidity, relevant positivity completeness, or relevant exemplification consistency yields a T2 countermodel in the same bounded family. The first $A1_L$ -drop witness is now rendered directly from the oracle as a two-node Kripke graph with complete local FDE tables, making both its accessibility relation and its positive/negative support matrix independently inspectable. This remains bounded indispensability, not global necessity.

Gate 8 demonstrates that the T2 bottleneck is definition-sensitive. The project-internal exact positive Godlikeness condition internalizes local positive reflection. Lean consequently proves

$$G\text{-exact}^+ + R^+ \Longrightarrow T2_{\text{exact}}^+$$

without $A1_L$, $COMP_P^G$, or $CONS_G^G$, while an executable model shows that exact positive Godlikeness remains compatible with genuine gluts.

The Anderson comparison shifts the modal work differently. Its literature-grounded positive interface classifies necessary properties of an individual through positivity and modifies essence accordingly. The project adds bilateral negative-evidence clauses and machine-checks their classical recovery. A complete-S5 model of the candidate has positive necessary Godlike existence while positive modal collapse fails for a contingent formula application. The Anderson T3 branch also reduces to symmetry alone, with a corresponding non-symmetric S4 countermodel. Scott and Anderson therefore share the same currently verified T3 return bridge while differing substantially in the upstream Godlikeness/essence mechanism.

The Fitting comparison exposes a third and more radical axis: the location of the property domain itself. The mechanized literature distinguishes intensions from extensions and applies Fitting positivity to extensions [4]. Enforcing that distinction in the four-valued setting first produced an instructive failure. If the extensional domain is identified with every bilateral support assignment, it contains a universal glut extension. Lean proves that the resulting unrestricted relevant-consistency condition excludes every positive Fitting-Godlike individual. The first unrestricted recovery implication was therefore logically correct but unsuitable as a substantive non-vacuous theorem.

The repaired candidate introduces a selected domain Adm of admissible extensions, closed under FDE negation but not globally constrained to consistency. On this domain Lean proves the non-vacuous Fitting essence theorem

$$\begin{aligned} & G\text{-admissible} + G_F\text{-realization} + A1_{L,F}^{adm} + REG_{G,F}^{adm} \\ & \implies G_F^+(x, w) \Rightarrow Ess_F^+(ext_w G, x, w), \end{aligned}$$

with no positivity-rigidity premise. This is a genuine structural difference from the current Scott-support recovery route: extensional necessary entailment operates on rigid extensions, so positivity need not be transported to each accessible world.

Gate 8 then minimizes this Fitting route further. Full relevant exemplification consistency is stronger than the A1-L proof itself requires: it is sufficient to demand consistency only when positivity completeness classifies the currently possessed extension negatively. Writing this targeted condition as $NegCons_{G,F}^{adm}$, Lean proves

$$\begin{aligned} & G\text{-admissible} + G_F\text{-realization} + A1_{L,F}^{adm} \\ & + COMP_{P,F}^{G,adm} + NegCons_{G,F}^{adm} \\ & \implies G_F^+(x, w) \Rightarrow Ess_F^+(ext_w G, x, w). \end{aligned}$$

A finite glut model separates the targeted condition from full consistency on the A1-L-only route. The weakening is itself A1-sensitive: restoring $A1_R$ allows Lean to reconstruct full relevant exemplification consistency from positivity completeness plus $NegCons_{G,F}^{adm}$. Thus the two A1 channels again have distinct structural roles even inside the Fitting comparison.

A second Fitting recovery route shows that positivity completeness is itself route-specific rather than globally necessary for the current essence/NE conclusion. The previously isolated Godlike-indiscernibility interface is now derived from a quotient-style condition on Adm . Let

$$x \approx_w^+ y$$

mean that every positively supported admissible extension contains x and y alike. Lean proves this to be an equivalence relation and proves that positive Fitting-Godlike individuals necessarily lie in the same equivalence class. If every admissible rigid extension is saturated with respect to these positive-profile classes, then

$$Sat_P^{adm} \implies Ind_G^{adm},$$

and therefore

$$G\text{-admissible} + G_F\text{-realization} + Sat_P^{adm} \implies G_F^+(x, w) \Rightarrow Ess_F^+(ext_w G, x, w)$$

without A1, $COMP_P^{G,adm}$, or exemplification consistency. A one-world three-entity finite model contains two distinct positive Godlike individuals in one non-singleton profile class, satisfies the saturation condition, and still has a relevant admissible extension with positivity value $\mathbf{N}$. Thus the classification gap remains real while the essence/NE route succeeds.

The quotient condition has a more canonical formulation than the initial indiscernibility interface suggested. FDE-negation closure upgrades positive-profile saturation to bilateral factorization, and quotient-respecting bilateral extensions are closed under the basic FDE operations. Ordinary algebra closure does not generate the quotient: a complete classical extension algebra can be closed under negation, conjunction, and disjunction while still containing a singleton that distinguishes two positivity-profile-equivalent individuals.

The entailment experiment sharpens this boundary further. The classical δ -filter closure described in the literature is closure of the family of positive extensions, not closure of the ambient property domain [4]. Naively imposing global entailment closure on Adm is too strong once FDE bottom is admissible, since bottom positively entails every extension. Lean proves

$$EntCl^{adm} + Adm(\perp_F) \implies \forall Y \text{ } Adm(Y),$$

and, with relevant exemplification consistency, this recreates the unrestricted-comprehension obstruction and excludes every positive Godlike witness. A

separate exhaustive finite fixture over all sixteen bilateral extensions on two entities satisfies $A2_F^+$, full domain-level entailment closure, FDE-algebra closure, and profile-compatible actual existence while profile saturation still fails. Thus entailment closure is not the missing source of the quotient.

Instead, Lean defines a canonical bilateral closure Sat_w that sends an arbitrary extension to the union of the positive-profile classes meeting its positive and negative support. This operation is extensive and monotone, idempotent up to bilateral extensional equivalence, and has the universal property of being the least quotient-respecting extension above its input. The earlier saturation principle is exactly the fixed-point condition

$$Sat_P^{adm} \iff \forall w, Y (Adm(Y) \rightarrow Sat_w(Y) \equiv Y).$$

Thus the selected Fitting property domain can be understood as living inside the fixed-point algebra of a canonical observational closure operator.

Actualist entailment introduces one additional interface. If actual existence is constant on positive-profile classes throughout the accessible region, then Lean proves

$$Entail_F^+(w, X, Y) \implies Entail_F^+(w, Sat_w(X), Sat_w(Y)).$$

A two-world finite countermodel shows that this descent can fail when two profile-equivalent individuals are split by the actual-existence predicate. The quotient analysis therefore reaches beyond property membership: actualist domain variation must also be coordinated with the entity quotient if Fitting entailment is to factor through it directly.

The quotient is now explicit in Lean. Entities at a source world are quotiented by positive admissible-property profile, and bilateral extensions are transported in both directions between quotient classes and profile-respecting entity extensions with round-trip equivalences. Moreover,

$$ExSat_P \iff \text{actual existence factors through every accessible source quotient.}$$

Under this exact representative-independence condition, quotient-level actualist entailment is equivalent in both directions to ordinary entailment on pullbacks. The bounded two-entity regression checks all 256 ordered pairs of saturated extensions and retains both **B** and **N**, so the construction is not a hidden return to bivalence.

The first filter analysis on this quotient gives a similarly sharp separation. In the FDE truth order, Lean defines proper delta-filters and their prime and non-adjunctive two-filter strengthenings. Point evaluation is a prime delta-filter: it designates **B**, but **N** is a self-negating gap outside the filter. Therefore primeness does not imply complement decision. The exhaustive two-point quotient audit finds four prime ordinary filters and all four are non-deciding; a separate ordinary-filter fixture has two positive-Godlike

points while local *COMP* fails for a possessed gap-bearing extension. The remaining question is which stronger filter interface, if any, contributes to essence/necessary existence without erasing this separation.

The literature's classical Fitting analysis supplies a useful contrast. Fitting positivity is a δ -ultrafilter on rigid extensions [4]. Isolating only its maximality component gives a complement-decision principle. In the bilateral reconstruction, Lean shows that complement decision together with relevant exemplification consistency already reconstructs local positive reflection and hence $COMP_P^{G,adm}$; complement decision together with $A1_R$ also reconstructs $COMP_P^{G,adm}$. The ultrafilter-style route therefore explains why classification is natural in the classical theory, while positive-profile saturation gives a genuinely different four-valued route that leaves relevant positivity gaps unresolved.

The admissible bilateral negative clauses also recover classically. Lean proves complementary recovery for extensional entailment, Godlikeness, essence, necessary actual exemplification, and necessary existence under classical coherence of the selected extensions. Thus the repair is not merely a positive-fragment construction.

The Fitting necessary-existence branch separates de re from de dicto explicitly. Freezing the current G -extension yields a frame-free theorem:

$$\text{admissible Fitting stack} + \Diamond_{de\,re}^+ \exists^E x \, ext_w G(x) \implies \Box_{de\,re}^+ \exists^E x \, ext_w G(x).$$

No reflexivity, symmetry, transitivity, seriality, or R^+ premise is used.

The earlier de-dicto bridge used full bilateral extension stability. The minimized proof reveals that only positive membership in G matters. Let $Pers_G^+$ preserve positive G -membership from the source of an accessibility edge to its target and let $RefI_G^+$ reflect positive membership back from target to source. Their conjunction, $STAB_G^+$, suffices for

$$\begin{aligned} &\text{minimized admissible Fitting stack} + STAB_G^+ + \Diamond^+ \exists^E x \, G_F(x) \\ &\implies \Box^+ \exists^E x \, G_F(x), \end{aligned}$$

again without an S4 or S5 frame premise. A complete-S5 finite fixture satisfies both A1 directions and even the older full regularity package while the positive G -extension is stable and the negative G -extension drifts. Positive necessary de-dicto Godlikeness still holds. Hence full bilateral $STAB_G$ is genuinely stronger than the positive theorem needs.

The positive stability assumption itself now has a derived form. Restrict forward positivity rigidity to admissible Fitting extensions as $R_{F,adm}^+$, and write $R_{F,adm}^{+\leftarrow}$ for converse transport of target positivity back to the source along an existing edge. Lean proves that the former derives $RefI_G^+$, while the latter derives $Pers_G^+$, so their conjunction yields all of $STAB_G^+$ without a frame condition. Symmetry plus $R_{F,adm}^+$ entails $R_{F,adm}^{+\leftarrow}$, recovering the older

route. A complete 256-case finite regression finds 112 candidates satisfying both transports, including 32 asymmetric candidates. Thus accessibility symmetry is strictly stronger than the bidirectional positivity transport consumed by the proof in this bounded family.

Finally, the admissible Fitting candidate admits the same qualitative anti-collapse phenomenon as the Anderson candidate. A complete two-world S5 model satisfies the encoded Fitting stack and positive necessary actual Godlikeness while a contingent $Q(a)$ refutes MC^+ . The admissible property domain in that model still contains genuine **B** information, so the result is not obtained by silently restoring global bivalence.

The comparative picture is therefore sharper than a hierarchy of stronger or weaker modal logics. Scott, Anderson, and Fitting distribute the decisive structure differently:

Scott support:	reflection recovery and R^+ precede $T2^+$,
Anderson:	necessary exemplification is built into Godlikeness,
Fitting:	positivity/essence range over selected rigid extensions, with classification and quotient-fixed-point recovery routes.

The accessibility relation matters, but it is not the only place where modal strength enters the theory.

The main open questions now separate accordingly. For Scott:

Which weaker principled four-valued assumptions, if any,
suffice for $T2^+$ globally?

For Fitting's quotient semantics:

Which four-valued filter principles on quotient fixed points
preserve gaps and gluts without reconstructing *COMP*?

For the ultrafilter comparison:

Can a genuinely four-valued δ -filter/ultrafilter be defined
on the closure fixed points without restoring *COMP*?

For the remaining de-dicto minimization:

Can $R_{F,adm}^{+\leftarrow}$ be restricted to G-triggered
or quotient-selected admissible extensions?

The modal generalization now has a precise first boundary. Every relational control model embeds exactly into the paired-neighborhood semantics, and complement duality suffices for classical modal recovery. In the complete

two-world local space there are twelve non-principal complement-dual pairs beyond the four principal relational pairs, while dropping complement duality permits modal **B** and **N** even for classical inputs. This establishes a genuine non-Kripke experimental region without yet claiming that the higher-order Scott, Anderson, or Fitting theorems transport unchanged.

Exact higher-order correspondence with the complete published formal developments, selected theorem-specific paired-neighborhood transports, and the remaining line-by-line source audit remain publication-stage requirements. The Gate-12 synthesis records this boundary explicitly. The paper continues to distinguish general Lean theorems, explicit finite countermodels, bounded exhaustive evidence, and open global-minimality claims.

A Formal Correspondence

Every claim in this manuscript of the form “Lean proves ...” refers to a named declaration in the accompanying development. This appendix gives the map. Lean modules are listed by file stem below

`formal/lean/Goedel4PEL/`; finite fixtures are listed by function name below `formal/finite/`. Both tracks are run in CI, and the Lean package is pinned to Lean 4.30.0. No declaration below uses `sorry`, an added axiom, or `native_decide`.

Premise names in the Lean signatures are the definitions of the same name in the corresponding module; where this manuscript abbreviates a premise list as a “stack”, the authoritative list is the Lean signature.

A.1 Kernel, modality, and the collapse channel

$MC^+ \leftrightarrow MC^-$ (Section 6)

`Analysis/Collapse.mcPlus_iff_mcMinus`

Bilateral box/diamond duality (Section 4)

`FourValued/PairedNeighborhood.diamondPlus_iff_boxNeg_neg,`
`.diamondNeg_iff_boxPlus_neg, .boxPlus_iff_diamondNeg_neg,`
`.boxNeg_iff_diamondPlus_neg`

Relational representation is definitional (Section 4)

`FourValued/PairedNeighborhood.boxPlus_ofRelation_iff` and the three further
`signed_ofRelation_iff` simp lemmas

Every relational pair is complement-dual (Section 4)

`FourValued/PairedNeighborhood.ofRelation_complementDual`

Complement duality gives classical modal recovery (Section 4)

`FourValued/PairedNeighborhood.boxNeg_iff_not_boxPlus_of_complementDual,`
`.diamondNeg_iff_not_diamondPlus_of_complementDual`

A.2 The Scott control chain

$A1_R + A2^+ \Rightarrow T1_T$ (Section 7)

GoedelScott/T1.t1T_of_a1R_a2Plus

Sufficient $T2^+$ recovery (Section 7)

GoedelScott/T2.t2Plus_of_regG

$T3^+$ from symmetry alone (Section 7)

GoedelScott/ScottFrames.t3Plus_of_possible_t2_a5_symmetric; the S5 presentation is .t3Plus_of_possible_t2_a5_s5_reduced and the unreduced form is GoedelScott/T3Collapse.t3Plus_of_possible_t2_a5

$T3^+ \Rightarrow GW$ on reflexive frames (Section 7)

GoedelScott/T3Collapse.gw_of_t3Plus

Essence-compressed collapse (Section 7)

GoedelScott/T3Collapse.formulaMCPlus_of_t2_t3_const

Bilateral classical recovery (Section 8.2)

Analysis/Recovery.postT_iff_pPos, .nentMinus_iff_not_nentPlus, .godMinus_iff_not_godPlus, .essMinus_iff_not_essPlus, .neMinus_iff_not_nePlus

A.3 Exact Godlikeness and the Anderson candidate

$G\text{-exact}^+ \Rightarrow G\text{-sup}^+$ (Section 8.2)

GoedelScott/GodlikeVariants.godExactPlus_implies_godPlus

Exact Godlikeness internalizes local reflection (Section 8.2)

GoedelScott/GodlikeVariants.reflection_of_godExactPlus

$G\text{-exact}^+ + R^+ \Rightarrow T2_{\text{exact}}^+$ (Section 8.2)

GoedelScott/GodlikeVariants.t2ExactPlus_of_rPlus

Anderson essence and necessary existence (Section 8.2)

GoedelScott/AndersonInterfaces.andersonGod_has_andersonEssence, .andersonGod_implies_boxGodExists, .andersonT3Plus_of_possibleGod

Anderson $T3$ from symmetry alone (Section 8.2)

GoedelScott/AndersonFrames.andersonT3Plus_of_possibleGod_symmetric; the S5 presentation is .andersonT3Plus_of_possibleGod_s5_reduced

Anderson bilateral classical recovery (Section 8.2)

GoedelScott/AndersonBilateral.andersonNecExMinus_iff_not_plus and the four further _iff_not_plus lemmas

A.4 Fitting: comprehension obstruction and admissible repair

Unrestricted consistency excludes every Godlike witness (Section 8.3)

GoedelScott/FittingSemantics.consGG_excludes_god, .regG_excludes_god

Admissible extensional essence theorem (Section 8.4)

GoedelScott/FittingAdmissible.god_has_currentExtension_essence_admissible; premises GAdmissible, GRealizationAdm, A1LAdm, RegGAdm

Admissible bilateral classical recovery (Section 8.4)

```
GoedelScott/FittingAdmissible.godMinusAdm_iff_not_godPlusAdm,  
.essMinusAdm_iff_not_essPlusAdm, .neMinusAdm_iff_not_nePlusAdm;  
GoedelScott/FittingAdmissibleRecovery. entailMinusAdm_iff_not_entailPlus,  
.boxExistsMinusAdm_iff_not_boxExistsPlus
```

A.5 Fitting: de re, de dicto, and the rigidity bridge

Frame-free de-re possibility to necessity (Section 8.5)

```
GoedelScott/FittingAdmissibleNecessaryExistence.possibleCurrentAdm_  
implies_necessary
```

De-dicto lifting under full extension stability (Section 8.5)

```
GoedelScott/FittingAdmissibleNecessaryExistence.possibleGodDeDictoAdm_  
implies_necessary
```

Bilateral stability implies positive stability (Section 8.7)

```
GoedelScott/FittingMinimality.gExtensionStableAlongRAdm_implies_  
gPosStable
```

Minimized de-dicto theorem (Section 8.7)

```
GoedelScott/FittingMinimality.possibleGodDeDictoAdm_implies_necessary_  
minimized
```

$R_{F,adm}^+ \Rightarrow Refl_G^+$ (Section 8.7)

```
GoedelScott/FittingRigidityBridge.rPlusAdm_implies_gPosReflects
```

$R_{F,adm}^{+\leftarrow} \Rightarrow Pers_G^+$ (Section 8.7)

```
GoedelScott/FittingRigidityBridge.rPlusConverseAdm_implies_gPosPersists
```

Symmetry entails converse transport (Section 8.7)

```
GoedelScott/FittingRigidityBridge.rPlusAdm_symmetric_implies_  
rPlusConverseAdm
```

Both transports give $STAB_G^+$ with no frame premise (Section 8.7)

```
GoedelScott/FittingRigidityBridge.rPlusAdm_rPlusConverseAdm_implies_  
gPosStable; the de-dicto conclusion is  
.possibleGodDeDictoAdm_implies_necessary_rPlus_bidirectional
```

A.6 Fitting: assumption minimization and the profile quotient

$CONS_{G,F}^{G,adm} \Rightarrow NegCons_{G,F}^{adm}$ (Section 8.7)

```
GoedelScott/FittingMinimality.consGGAdm_implies_negClassConsistencyAdm
```

Essence theorem on the targeted package (Section 8.7)

```
GoedelScott/FittingMinimality.god_has_currentExtension_essence_negClass
```

Al_R reconstructs full consistency (Section 8.7)

```
GoedelScott/FittingMinimalityInteractions.a1R_regGNegClassAdm_implies_  
consGGAdm, .a1R_regGNegClassAdm_implies_regGAdm
```

Classification-free indiscernibility route (Section 8.7)

```
GoedelScott/FittingAlternativeRecovery.god_has_currentExtension_essence_  
indiscernible; the necessary-existence continuations are
```

`.possibleCurrentAdm_implies_necessary_indiscernible` and
`.possibleGodDeDictoAdm_implies_necessary_indiscernible`
 $\approx_w^+$ is an equivalence relation (Section 8.7)
`GoedelScott/FittingAdmissibleStructure.positiveProfileEqAdm_refl,`
`.positiveProfileEqAdm_symm, .positiveProfileEqAdm_trans`
Godlike individuals share a profile class (Section 8.7)
`GoedelScott/FittingAdmissibleStructure.godlike_implies_`
`positiveProfileEqAdm`
 $Sat_P^{adm} \Rightarrow Ind_G^{adm}$ (Section 8.7)
`GoedelScott/FittingAdmissibleStructure.positiveProfileSaturated_implies_`
`godlikeIndiscernibility`
 $Sat_P^{adm} \Leftrightarrow Sat_{\pm}^{adm}$ (Sections 8.7 and 8.8)
`GoedelScott/FittingAdmissibleStructure.positiveProfileSaturated_iff_`
`bilateralProfileSaturated`
Complement decision reconstructs COMP (Section 8.7)
`GoedelScott/FittingAdmissibleStructure.complementDecision_cons_implies_`
`compPGAdm, .complementDecision_aIR_implies_compPGAdm`; the intermediate
reflection step is `.complementDecision_cons_implies_positiveReflection`

A.7 Fitting: quotient-domain algebra

Quotient factorization is closed under the FDE operations (Section 8.8)
`GoedelScott/FittingAdmissibleAlgebra.respectsProfileAt_negate,`
`.respectsProfileAt_fdeConj, .respectsProfileAt_fdeDisj,`
`.respectsProfileAt_fdeTop, .respectsProfileAt_fdeBot,` collected as
`.profileFactorization_closed_under_fde`
Admissible extensions respect the quotient under Sat_P^{adm} (Section 8.8)
`GoedelScott/FittingAdmissibleAlgebra.admissible_respectsProfileAt`

A.8 Fitting: entailment, the explicit quotient, and filters

Domain entailment closure forces unrestricted comprehension (Section 8.9)
`GoedelScott/FittingEntailmentQuotient.globalEntailmentClosed_bot_implies_`
`unrestricted; with relevant consistency,`
`.globalEntailmentClosed_bot_cons_excludes_god`
 Sat_w is a closure operator (Section 8.9)
`GoedelScott/FittingEntailmentQuotient.profileSaturateAt_extensive,`
`.profileSaturateAt_mono, .profileSaturateAt_idempotent,`
`.profileSaturateAt_respectsProfile`
Least quotient-respecting hull (Section 8.9)
`GoedelScott/FittingEntailmentQuotient.profileSaturateAt_least_respecting`
Saturation is the fixed-point condition (Section 8.9)
`GoedelScott/FittingEntailmentQuotient.positiveProfileSaturated_iff_`
`fixedByProfileClosure`
Entailment descends through Sat_w (Section 8.9)
`GoedelScott/FittingEntailmentQuotient.entailPlus_profileSaturateAt`

Push/pull round trips on the quotient (Section 8.9)

```
GoedelScott/FittingActualistQuotient.from_toProfileQuotient_equivalent,  
.to_fromProfileQuotient_equivalent,  
.respectingExtension_has_quotientRepresentation,  
.profileSaturateAt_has_quotientRepresentation
```

ExSat_P is exactly existence factorization (Section 8.9)

```
GoedelScott/FittingActualistQuotient.profileExistenceSaturated_iff_  
factors
```

Quotient entailment equals entailment on pullbacks (Section 8.9)

```
GoedelScott/FittingActualistQuotient.profileQuotientEntailPlus_iff_  
pullback, .entailPlus_iff_profileQuotient
```

Point evaluation is a non-deciding prime filter (Section 8.9.1)

```
GoedelScott/FittingQuotientDeltaFilter.evaluationFilter_primeDeltaFilter,  
.evaluationFilter_designates_both, .evaluationFilter_neither_undecided,  
.evaluationFilter_not_complementDeciding,  
.profileQuotient_has_nondeciding_primeDeltaFilter
```

Truth order reduces to positive inclusion classically (Section 8.9.1)

```
GoedelScott/FittingQuotientDeltaFilter.truthLE_iff_posSupportLE_of_  
classical
```

Delta-filters satisfy the two-filter law (Section 8.9.1)

```
GoedelScott/FittingQuotientDeltaFilter.positiveDeltaFilter_implies_  
twoFilter
```

FDE operations are meet and join for $\leq_t$ (Section 8.10)

```
GoedelScott/FittingDeltaFilter.fdeConj_truthLe_left,  
.fdeConj_truthLe_right, .truthLe_fdeConj, .left_truthLe_fdeDisj,  
.right_truthLe_fdeDisj, .fdeDisj_truthLe; negation reverses the order by  
.extensionTruthLe_negate
```

Classical recovery of the filter notion (Section 8.10)

```
GoedelScott/FittingDeltaFilter.deltaFilter_iff_classicalDeltaFilter_of_  
classical
```

Carrier extensions are closure fixed points (Section 8.10)

```
GoedelScott/FittingDeltaFilter.profileDeltaFilter_admissible_fixed
```

Polarity gives both A1 directions and the dual ideal (Section 8.10)

```
GoedelScott/FittingDeltaFilter.bilateralProfileDeltaFilter_implies_a1L,  
.bilateralProfileDeltaFilter_implies_a1R,  
.bilateralProfileDeltaFilter_bot_negative,  
.bilateralProfileDeltaFilter_top_not_negative,  
.bilateralProfileDeltaFilter_negative_downward,  
.bilateralProfileDeltaFilter_negative_disj
```

Ultrafilter maximality reconstructs COMP (Section 8.10)

```
GoedelScott/FittingDeltaFilter.profileDeltaUltrafilter_cons_implies_  
compPGAdm, .profileDeltaUltrafilter_a1R_implies_compPGAdm. Both consume  
only the maximal field, via  
.profileDeltaUltrafilter_implies_complementDecision
```

A.9 Finite fixtures and bounded searches

All entries are functions in `formal/finite/`, executed by the regression suite `test_checker.py`.

Glut and gap $T2^+$ countermodels; 204-model control search

```
checker.t2_glut_countermodel, .t2_gap_countermodel,  
.exhaustive_reg_g_implies_t2,  
.exhaustive_mc_schema_equivalence_two_world_s5
```

873-model bounded $T2$ assumption search; unrestricted $T1$ obstruction

```
gate7_search.exhaustive_t2_assumption_minimality,  
.validate_t1_glut_countermodel. The family fixes  $P(G) = \mathbf{T}$  and  $P(\neg G) = \mathbf{F}$  at  
both worlds
```

Anderson and Scott $S4$ countermodels; no-collapse fixture

```
gate8_anderson.validate_anderson_no_collapse_model,  
gate8_frames.validate_anderson_s4_countermodel,  
gate8_scott_frames.validate_scott_s4_countermodel
```

Fitting de-re/de-dicto separation; necessary God without collapse

```
gate8_fitting.validate_fitting_admissible_de_re_de_dicto_model, gate8_  
fitting_no_collapse.validate_fitting_necessary_god_no_collapse_model
```

Targeted consistency and positive-only stability strictness

```
gate8_fitting_minimality.validate_neg_class_consistency_strictness,  
.validate_positive_stability_strictness
```

COMP-free indiscernibility and profile-saturation fixtures

```
gate8_fitting_comp.validate_indiscernibility_route_without_comp, gate8_  
fitting_profile_saturation.validate_profile_saturation_route_without_comp
```

FDE algebra closure does not force saturation

```
gate8_fitting_algebra.validate_fde_algebra_does_not_force_profile_  
saturation
```

Entailment closure and existence-split fixtures

```
gate8_fitting_ entailment.validate_ entailment_closure_does_not_force_  
profile_saturation,  
.validate_existence_saturation_is_needed_for_ entailment_descent
```

Rigidity bridge, 256 configurations

```
gate8_fitting_rigidity_bridge.validate_rigidity_bridge_exhaustive and its  
two separation fixtures;  
gate8_fitting_converse_rigidity.validate_converse_rigidity_exhaustive  
returns the tuple (256, 152, 112, 80, 32)
```

Explicit actualist quotient, 16 extensions and 256 entailment pairs

```
gate8_fitting_actualist_quotient.validate_actualist_profile_quotient,  
.validate_actualist_quotient_requires_existence_factorization
```

Two-point filter audit (166/15/4/112/10)

```
gate10_quotient_delta_filters.validate_quotient_delta_filters
```

Profile δ -filter without maximality or COMP

```
gate8_fitting_delta_filter.validate_delta_filter_does_not_restore_comp
```

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